A Comprehensive Study and Performance Evaluation of Indoor Positioning Methods in BLE and UWB Technologies

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***Abstract*—Location awareness is a crucial aspect in making things more intelligent and secure. It offers context-aware services in indoor positioning systems (IPSs), such as navigation assistance in airports, shopping malls, hospitals, and asset tracking in warehouses. The exchange of data in an Internet of Things (IoT)-based network with location awareness enables real-time tracking in IPS, which is one of the rapidly enhancing research domains. Bluetooth low energy (BLE) and ultrawideband (UWB) stand as the most widely recognized technologies in IPS applications by presenting a wide range of positioning methods. This article conducted a comprehensive study of BLE and UWB technologies and performance evaluation of their positioning methods, in consideration to their accuracy, reliability, and implementation complexity in an indoor environment. The article is mainly focused on received signal strength indicator (RSSI), angle of arrival (AoA), angle of departure (AoD), phase-based ranging (PBR), round-trip time (RTT), time of flight (ToF), and time difference of arrival (TDoA). The evaluation of these methods is classified into three categories such as signal strength, angle, and time-based positioning methods under the line-of-sight (LoS) and nonline-of-sight (NLoS) conditions. The positioning accuracy of real-time applications can vary depending on the type of IPS technology, such as wireless fidelity (Wi-Fi), BLE, UWB, and radio frequency identification (RFID), and their positioning methods, respectively. The choice of a specific IPS technology and its implementation strategy depends on an application's requirements and use case. BLE and UWB are considered the most reliable and adaptive IPS technologies due to their characteristics, e.g., positioning accuracy, low-power consumption, implementation cost, and support in modern devices like smartphones, tablets, and laptops. To the best of our knowledge, this is the first head-to-head technical comparison between BLE and UWB technologies and evaluation of their positioning methods have ever conducted in consideration of their increasing demands in IPS applications.**

***Index Terms*—Bluetooth low energy (BLE), indoor positioning system (IPS), Internet of Things (IoT), ultrawideband (UWB).**

I. Introduction

THE rapid growth of the Internet of Things (IoT) and the demand for indoor positioning systems (IPSs) to offer precise positioning information have been remarkably increasing [1]. Traditional outdoor positioning systems (OPSs) like Global Navigation Satellite Systems (GNSS) may not provide reliable performance in indoor location due to multipath effects, signal attenuation, and lack of line-of-sight (LoS) with satellites [2]. An IPS technology plays a monumental role in providing location awareness and security, which can facilitate a hassle-free user experience in an indoor environment [3].

Localization and positioning are the most common terms in IPS, usually used interchangeably depending on the status of an object in an indoor location. Localization covers a broader concept, involving the determination of an object's position and its relationship to the surrounding environment, e.g., the position of an object inside a room or building. Positioning, on the other hand, refers to the process of determining an object's exact coordinates in a given space, which is primarily used in the context of navigation and tracking [4].

Over the last two decades, the advancements of different IPS technologies in providing enhanced positioning accuracy have been remarkable, e.g., wireless fidelity (Wi-Fi), Bluetooth low energy (BLE), ultrawideband (UWB), ZigBee, long range (LoRa), Sigfox, narrowband IoT (NB-IoT), visible light communication, radio frequency identification (RFID), inertial systems, and cellular technology [4], [5], [6]. These technologies offer services based on the need and demand of different IPS applications, and their utilization in IPS depends on a number of different parameters which needs to be considered during its implementation, e.g., complexity of indoor location, energy consumption, implementation cost, positioning accuracy, communication range and their reliability to LoS, and nonline-of-sight (NLoS) conditions [7].

IoT-based applications are mainly focused on energy-efficient technologies [8]. Although most of the IPS technologies are energy efficient, positioning accuracy remains a major flaw in most of them, e.g., Wi-Fi, Zigbee, and LoRa. Meanwhile, BLE and UWB technologies not only consume low energy among other IPS technologies, they also offer precise positioning information and are highly adaptive to indoor environments, which can be found in almost all modern devices, e.g., smartphones, tablets, and computers [9], [10].

The complexity of an IPS application is proportional to its positioning method. Both BLE and UWB technologies offer a large number of distinct positioning methods, which can be classified into three categories: the signal strength-based positioning methods (SSPMs), which determine the position of an object by analyzing the strength of wireless signals

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received from known transmitters [11], the signal angle-based positioning methods (SAPMs), which determine the position of an object based on the angle of received or transmitted signal [1], and the signal time-based positioning methods (STPMs), which determines the position of an object based on the time signal takes to reach different receivers [12].

In accordance with the given classification, BLE technology comparatively has a large number of positioning methods utilized in each classified category, e.g., received signal strength indicator (RSSI) in SSPMs, angle of arrival (AoA), and angle of departure (AoD) in SAPMs, round-trip time (RTT), and phase-based ranging (PBR), in STPMs. While UWB, on the other hand, has been known for STPMs mainly time of flight (ToF), and time difference of arrival (TDoA). Evaluation of RSSI, AoA, AoD, RTT, and PBR in this study is based on BLE technology, whereas the ToF and TDoA methods are analyzed based on UWB technology.

While these positioning methods have been individually investigated in various research articles, the studies show that integration of BLE's and UWB's positioning methods can provide even higher accuracy, scalability, and robustness, which can open up new possibilities for large-scale deployment in different environments, including complex, dynamic, and littered indoor spaces [13].

This article has conducted an extensive study of BLE and UWB technologies and performance evaluation of their positioning methods, in accordance with their accuracy, reliability, cost, and implementation complexity in an indoor location. Each method is furthermore investigated based on its strengths and limitations in different operational contexts by addressing the challenges it poses. Toward the end of this article, we have proposed different optimization strategies to enhance the estimation accuracy of BLE's and UWB's positioning methods, specifically while dealing with NLoS environments.

Ultimately, this research aims to provide an in-depth understanding of the current state of the art of BLE and UBW technologies and their positioning methods. This is in accordance with the needs for precise positioning information in IPS applications. Their evaluation is further corroborated by a simulation using MATLAB software. To the best of our knowledge, this is the first head-to-head technical comparison between BLE's and UWB's positioning methods ever conducted.

The key contributions of this article are as follows.

1) We have conducted a comprehensive study of BLE and UWB technologies, focusing on their utilization for IPS applications, by examining their fundamental characteristics, operational principles, and limitations to assess their reliability for accurate positioning.
2) A detailed performance evaluation of BLE's and UWB's positioning methods was carried out under LoS and NLoS conditions, considering their essential factors, e.g., positioning accuracy, implementation complexity, and security aspects, by analyzing how each technology performs in varying indoor conditions, and highlighting the fitness of BLE and UWB in different scenarios.
3) In addition, we presented a detailed analysis of BLE's STPMs (channel sounding (CS) techniques), e.g., RTT and PBR, with particular emphasis on their performance under LoS and NLoS conditions. The results were benchmarked against UWB's STPMs. This comparative study offers new insights into the relative strengths, weaknesses, and practical limitations of SSPMs, SAPMs, and STPMs in both technologies.
4) Finally, we proposed optimization strategies to address the inherent challenges associated with both technologies, especially under NLoS conditions where performance tends to degrade. This article addresses the challenges faced by BLE and UWB systems in real-world indoor environments associated with IPS deployments, and practical recommendations are given to enhance their reliability and effectiveness in IPS implementations.

In general, a classified approach in BLEs and UWBs' positioning methods can provide a better understanding of their technical aspect. This can represent a clear survey structure. However, there is no prior work that has conducted a classification-based approach in BLE and UWB technologies. The main motive of this research is to provide a new contribution addressing the gap in previously surveyed articles.

The rest of this article is structured as follows. Section II explores related work on BLE's and UWB's positioning methods. Section III demonstrates BLE and UWB technologies evolution in location-based services (LBSs). Section IV explores BLEs and UWBs' positioning methods. Section V represents common challenges in BLE and UWB and explores optimization algorithms. The conclusion is presented in Section VI.

II. Literature Survey

This section undertakes a systematic literature review centered on BLE and UWB technologies, with a particular emphasis on their respective positioning methods. The selection of reviewed articles is based on rigorous technical criteria, including the evaluation of their positioning accuracy, complexity, computational efficiency, and applicability across diverse use cases. This encompasses a classified positioning category, with special attention given to studies integrating a broad spectrum of IPS technologies. Priority is accorded to publications that either exclusively investigate BLE or UWB technologies and their positioning methods or provide comparative insights with alternative IPS technologies (see [14], [12], [15], [16], [17]). This approach ensures a critical synthesis of current advancements, addressing both the theoretical underpinnings and their practical deployment.

A. General Reviews on BLE and UWB

A state-of-the-art study in [18] evaluates the BLE-based positioning accuracy metric by classifying BLE's positioning methods into ten major types. This includes approaches such as AoA, fingerprinting, fusion, centroid, filtration, logarithmic path-loss model (LPM), trilateration, machine learning (ML), and optimization methods. The article, however, has explored BLE's integrated approach for positioning, but its fundamental methods have not been evaluated. Yang et al. [19] made a

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comprehensive study on Bluetooth technology exploring its utilization in different IPS applications and current solutions from academia and industry. The major part of the article is a theoretical explanation of Bluetooth technology. Zhuang et al. [13] presented a detailed analysis of Bluetooth's positioning accuracy, working principles, and future trends. However, Morgan [18], Yang et al. [19], and Zhuang et al. [13] have an illustrative approach to Bluetooth; there has not been any technical overview specific to Bluetooth's positioning methods, which could reveal their potential power and contribution to the field of IPS.

Liu et al. [20] conducted a study of UWB IEEE 802.15.4z and IEEE 802.15.4 standards focusing on performance, accuracy, and their energy efficiency in accordance with NoS and NLoS conditions. Their obtained results indicate that both the standards achieve centimeter-level ranging accuracy, where IEEE 802.15.4z offers better performance in NLoS conditions in comparison to the IEEE 802.15.4 standard.

There are very few articles specifically focused on BLE and UWB technologies and their positioning methods (see [13], [18], [19], [20]). Most of the available studies present a generalized analysis of IPS technologies and their positioning methods. Asaad and Maghdid [3] conducted a survey on different IPS technologies and their positioning methods within the domain of IoT by discussing their use case scenarios. The article, however, has explored different aspects of IPS technologies, yet their technical evaluations and the relevant challenges that these technologies might pose are not addressed. Similarly, Zafari et al. [21] made a study on different IPS technologies, mainly focused on Wi-Fi, RFID, UWB, and Bluetooth technologies by exploring their positioning methods, e.g., AoA, ToF, RToF, and RSS. Their study consists of cost, computational and time complexity, power consumption, scalability, reliability, and security parameters. Hayward et al. [22] reviewed IPS methods and their applications, particularly within industry contexts. The authors explored the benefits and limitations of different proposed technologies, considering their usefulness in industry-based applications. And Sartayeva et al. [23] presented a six-layer framework model for IPS. It consists of application, method, data, network, communication, and device. The article is mainly focused on ML-based approaches particularly in COVID-19-related applications. The study lacks key positioning methods to be proposed for precise positioning accuracy, and technical details of the evaluated methods have also been neglected.

B. Signal Strength-Based Positioning Methods

SSPMs are among the most commonly used techniques, which is used to estimate the position of an object by analyzing the strength of radio signals, e.g., RSSI. Its implication is adopted by a huge number of IPS technologies due to its implementation simplicity. The RSSI method is mostly hybrid in order to provide enhanced positioning accuracy, e.g., the integration of particle filter (PF) algorithm in [24], which utilizes a BLE-based RSSI method using the smartphones as mobile nodes with a static BLE sniffers receivers for collecting and recording the signals received from the mobile node. Assayag et al. [11] used an adaptive RSSI-based model to select the best anchor nodes for the long-distance signal coverage using a data fusion technique. The proposed model does not require training or prior knowledge of a device position.

RSSI has also been used in other positioning techniques, for instance, the fingerprint method, which is based on RSS values and mainly has two-step implications. In the first stage, we need to collect the RSS signals, and in the second stage, we identify the location of an object using the precollected data. This method has also been used in a number of different IPS applications. Duong and Dinh [25] proposed a fingerprint positioning technique based on k-nearest neighbors (kNNs) to decide the position of an object. The main issue with the fingerprint method is that it always relies on the precollected data, and it is a location-oriented approach. In case of changing or extending the location, we need to recollect the RSSI data and store it in the database. The fingerprint method mostly has poor positioning accuracy as compared with other methods [26].

C. Signal Angle-Based Positioning Methods

In SAPMs, e.g., AoA and AoD, designing an antenna array is one of the concerns that can cause additional implementation costs. SAPMs carry a unique feature, which is the ability to identify the direction from which signals are coming and the ability of beamforming, through which the signal can be guided to a specific direction, which eventually enables the signal to travel over long distances.

A one-directional AoA considering the challenges in a 3-D-based method—which can be problematic in some limited indoor locations—is proposed in [27]. The proposed technique is based on the angle between the array orientation and the ray pointing to the target, which can be directly obtained by each sensor by formulating a nonlinear least squares problem. The 1-D-based approach in some environments offers better results; however, the issue it poses is the increase in the number of iterations based on the involvement of multiple locators.

While the traditional IPS applications require a threefold communication architecture, such as reference nodes having predefined positions, a mobile node with an unknown position, and a positioning engine that identifies the mobile node's position. However, there is an alternative approach called device-free localization (DFL), which is based on detecting and tracking the passive targets without requiring the mobile node. Hong et al. [28] proposed a DFL-based two-stage learning AoA method to achieve high positioning accuracy, which can be efficient through exploiting multipath components (MPCs) in UWB systems. The authors have designed a passive localization net (PLN) to improve mobile node identification and the proximity performance of MPC, where reidentification of MPC is performed in the first stage based on the coarse preidentified MPC. In the second stage, the original AoA of the MPC is refined, using the least squares estimator (LSE), in accordance with the first stage. While the DFL approach is cost-efficient, it can also be vulnerable based on the environment and the mobile nodes' movement.

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Han et al. [9] explored the AoD method in UWB's latest IEEE 802.15.4z standard based on two switched array antennas utilizing Qorvo's DW3120 UWB IC, and NRF52840 Nordic Microprocessor. In addition to BLE and UWB, the AoD method has been used in other technologies due to its potential strength in indoor-based navigation. Wang et al. [29] used AoD in a wireless sensor network (WSN), where a rotary antenna is used to create a synthetic uniform circular array (SUCA). This allows for the distributed positioning of nodes in a multipath location. Both AoA and AoD can obtain high positioning accuracy in an indoor location due to their support for antenna arrays in the reference node that can perform beamforming, resulting in a strong signal strength.

D. Signal Time-Based Positioning Methods

UWB technology is famous for STPMs, which determine an object's position by analyzing the time it takes for a signal to travel between the mobile node and reference nodes, e.g., ToF and TDoA methods. On the other hand, BLE technology has adopted STPMs in 2024, by introducing the concept of CS, which consists of RTT and PBR methods. However, the implementation of BLE's CS methods is more complicated.

Mazraani et al. [12] proposed a combined TDoA and ToF approach, stating that it can overcome the issue of power consumption caused by the ToF method using the concept of TDoA, while maintaining the same accuracy as the ToF method with a reduction of 70% power consumption. Wang et al. [30] utilized the RTT method based on its two-way ranging mechanism. The article focuses on an asynchronous ranging scheme for low-frequency oscillators with inherent drift, which provides better positioning accuracy and stability. Measuring and reporting each node's response delay is based on multiple iterations of the symmetric double-sided two-way ranging (SDS-TWR) method. It can reduce ranging errors in the device's oscillator and variations in the response time of a node due to the timing resolution of the sampling clock and timing drift. Although CS methods have newly been adopted in BLE, they have been utilized in different types of applications. Zand et al. [31] proposed a BLE-based PBR solution to calculate the range between devices with the ability to mitigate the multipath fading problem. This approach uses channel hopping (CH) to exchange tones in the entire 2.4-GHz frequency band, to overcome slow CH and frame spacing between two connected devices, which affects ranging accuracy. The article has introduced a mathematical model for analyzing the impact of the BLE link layer protocol on reducing ranging errors.

While STPMs provide precise positioning accuracy, on the other hand, they are more vulnerable to NLoS conditions due to their synchronization (SYNC) issue [32]; meanwhile, there have been a number of different approaches proposed to address this issue. For instance, Chen et al. [33] proposed the RefLoc method, using UWB technology, utilizing the DS-TWR method, which is based on ToF, using the idea of RefLoc. This is used in the UWB positioning approach by leveraging its unique benefits to address the NLoS problem and identify links among reference nodes with known positions, and then using those links as references to identify the position of a mobile node.

Wang et al. [34] explored the concept of DFL using the ToF method. While the proposed ToF-based DFL approach can achieve higher positioning accuracy, its implementation is more complex due to a sequential pairwise ToF measurement for each pair of nodes. This makes it inefficient in time and energy-sensitive applications. The proposed approach is based on merging the link state prediction function into a PF framework in order to predict the informative shadowed links by using a parallel scanning scheme for acquiring multiple pairs of ToF measurements simultaneously to reduce the time and energy consumption of the traditional ToF approach.

Based on our analysis, the most commonly used BLE and UWB positioning methods are RSSI, AoA, ToF, TDoA, and RTT, as we can find a large number of remarkable publications.

Existing literature reviews on BLE and UWB technologies, and their respective positioning methods, provide limited information. The comparison between this article and the gap in the related works is represented in Table I, although most of these articles have a theoretical approach as their review methodologies are lacking technical knowledge, which is highly required as the state of the art in BLE and UWB.

TABLE I
SUMMARY OF STATE-OF-THE-ART OF BLE AND UWB REVIEW ARTICLES

Positioning Technology			SSPMs	SAPMs		STPMs			
Ref.	BLE	UWB	RSSI	AoA	AoD	RTT	PBR	TDoA	ToF
[18]	✓	✗	✓	✓	✗	✗	✗	✗	✗
[19]	✓	✗	✓	✗	✗	✗	✗	✗	✗
[13]	✓	✓	✓	✓	✗	✗	✗	✓	✗
[20]	✗	✓	✗	✗	✗	✓	✗	✗	✓
[3]	✗	✓	✓	✓	✗	✓	✗	✓	✓
[21]	✓	✓	✓	✓	✗	✗	✗	✓	✗
[22]	✓	✓	✓	✓	✗	✗	✗	✓	✗
[23]	✓	✓	✓	✓	✓	✗	✗	✓	✗
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓

III. LBS EVOLUTIONS IN BLE AND UWB TECHNOLOGIES

IPS technologies have evolved significantly since their inception in the late 20th century. They were initially developed for military and industrial applications. The early systems were mainly used for asset tracking in large warehouses and manufacturing facilities. The evolution of IPS continued with significant advancements in wireless communication technologies, particularly Bluetooth and UWB. In the early 2000s, Bluetooth emerged as a cost-efficient and widely adopted solution for short-range communication. This enabled Bluetooth-based IPS application to be more accessible and scalable, especially with the introduction of BLE in 2010 [13].

However, while Bluetooth provided good coverage and low-power consumption, its precision for accurately tracking an object was not satisfactory. On the other hand, the introduction of UWB technology in the mid-2000s offered enhanced precision. This allowed for submeter accuracy in real-time tracking, especially in challenging environments [13]. Over time, both technologies have been remarkably improved by offering techniques that can estimate the position of an object up to centimeter-level accuracy. BLE and UWB are considered

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the most reliable and adaptive IPS technologies utilized by a huge number of IPS applications. They are both focused on low-energy consumption and accurate positioning estimation. In this section, we explore both technologies, their advancement throughout time, and capabilities in the realm of wireless communication individually [35]. However, they have a distinct nature of communication, but in terms of LBSs, they offer great results.

A. BLE Technology

Bluetooth technology has been offering services to a number of different applications for decades. The special interest group (SIG) collaborated on Bluetooth in 1998, and in the following year, the specification of Bluetooth 1.0 was released, followed by Bluetooth 1.2, 2.0, and 3.0. These series of Bluetooth are known as classic Bluetooth operating in 2.4-GHz industrial scientific and medical (ISM) band, using 79 radio frequency (RF) channels with 1 MHz spacing from 2.402 to 2.480 GHz [36].

Bluetooth technology has evolved significantly in the context of IPS over the years. It was initially adopted in the early 2010 with the concept of BLE as Bluetooth 4.0. This was a major breakthrough in Bluetooth technology with transmission power from 0.01 to 10 MW, which is much less than classic Bluetooth. In BLE, the RF channels were reduced from 79 to 40 channels each at 2.4 GHz, which was referred to as Bluetooth Smart [37]. A part of BLE 4.0 was for novel applications in healthcare [38], fitness, and security [39].

Bluetooth kept introducing new features since then, e.g., BLE 5.0 in 2016 included faster data transfer speeds of up to 2 MB/s, while BLE 4.0 had a maximum data transfer rate of 1 MB/s, better power efficiency, increased communication range up to 300 m in BLE beacon devices, and offered robust support for IoT devices [40]. However, BLE was providing different services, including LBS using SSPMs, yet its precision in identifying the position of an object could not meet the expectations and demands of IPS applications.

In 2019, Bluetooth technology released the BLE 5.1 core specification, which introduced the concept of direction finding (DF) feature that consists of AoA and AoD methods. This was a pivotal advancement in BLE since its first release in 2016 and has enabled it to provide up to submeter-level positioning precision [1]. This led to widespread adoption of BLE in different industries, including real-time locating systems (RTLSs) for asset tracking in warehouses and hospitals, and navigation systems in shopping malls and airports. BLE 5.2 and 5.3 in 2023 further integrated with technologies like augmented reality (AR) and virtual reality (VR), opening a new era of possibilities for immersive user experiences.

However, BLE provides good coverage range, low-power consumption, and the ability to identify the direction of signals using DF features. Its ability to provide higher positioning accuracy could not meet the required precision standards to compete with UWB technology. This issue was addressed in 2024 by introducing the concept of CS, which consists of PBR and RTT methods in BLE's 6.0 version. It opened a new era in the history of Bluetooth technology by providing enhanced positioning precision, allowing for subcentimeter-level accuracy. This was BLE's first attempt for STPMs, with an improved security feature that advances the use case of BLE technology in IPS. Hence, Bluetooth became a true competitor of UWB technology in both precise positioning estimation and security [13].

TABLE II
COMPARISON OF BLUETOOTH VERSIONS FOR LBS

BLE Versions	Added Features and Specifications
BLE 4.0 (2010)	• Introduced Bluetooth Low Energy (BLE) • Basic proximity detection using RSSI • Support for IPv6 and IPSP • Improved coexistence with other technologies • Transmission range up-to $100m$ and speed $1mbps$
BLE 5.0 (2016)	• Introduced "LE Coded" that can quadruple the range • Improved RSSI for better proximity estimation • Support for mesh networking • Transmission range $\approx 240m$ and speed upto $2Mbps$.
BLE 5.1 (2019)	• Improved proximity estimation up-to 1 meter • Introduced AoA and AoD for direction finding • Support for GATT caching • Encryption, periodic advertising with reposes (PAwR)
BLE 6.0 (2024)	• Introduced Channel Sounding (PBR and RTT) • Support of Connection-Oriented Transmission • Support for multiple device connections • Enhanced mesh networking • Support for IoT devices with multi-stream audio.

CS methods enable BLE technology with a two-way ranging approach between two BLE devices, which are referred to as initiator and reflector devices. These methods offer improved distance measurement accuracy up to centimeter-level in applications such as asset trackers and smart locks [41]. BLE 6.0 also offers a new physical layer (PHY) that uses amplitude-shift keying modulation (ASKM) across 72 RF channels instead of the traditional 40 RF channels, which is incorporated into its stack's link layer to handle the setup, timing, and security of CS features. A summary of different BLE versions and their advancements in LBS over time is represented in Table II.

These advancements have significantly improved BLE's capabilities, making it a more effective, secure, and suitable option for IPS with a wide range of applications, including IoT devices, audio streaming, and precise positioning tracking.

BLE technology is well known for its low-energy consumption, which is used in resource-constrained sensor nodes, particularly in IPS applications [42], which is popular due to its cost-efficient implementation by eliminating the need for additional hardware [39]. Some of the known applications using BLE apart from positioning estimation are connecting wireless sensors, healthcare [38], smart homes, vehicular ad hoc networks, multimedia streaming, and mobile payment systems [13]. This distinct use case of BLE technology enables Bluetooth to provide different chipsets, such as classic Bluetooth chipsets used for audio streaming, dual-mode chipsets used in applications like fitness watches, and BLE chipsets used in applications like smart parking systems. The dual-mode chipset aimed to support low-energy basic rate (BR) and enhanced data rate (EDR) communications commonly used in

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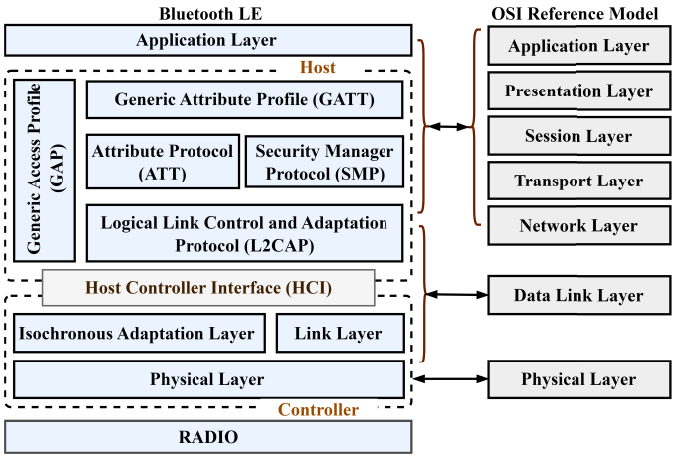


Fig. 1. BLE stack layer with open systems intercommunication (OSI) reference model [41].

smartphones and tablets. Over time, it was replaced with a BLE chipset starting from BLE 4.2 in most modern devices [10], [42].

Communication is an essential part of IPS technologies, which is related to the structured software framework and the frame structure that facilitates communication between devices by using short-wavelength, ultrahigh frequency (UHF) radio waves over a short range to form a personal area network (PAN). The stack architecture in BLE is explored through layers, which consist of two main parts: the controller side and the host side. Controller consists of PHY layer, link layer, and isochronous adaptation layer (IAL). The host side contains logical link control adaptation protocol (L2CAP), generic access profile (GAP), attribute protocol (ATT), security manager protocol (SMP), and generic attribute profile (GATT). The flow of communication between these layers is illustrated in Fig. 1.

Communication between controller and host is managed through the host controller interface (HCI), and the encapsulated data packets between the lower layers and upper layers are handled by L2CAP. ATT manages to define the attributes, e.g., basic units of data in the BLE database, where each attribute has a type which is identified by a universally unique identifier (UUID), a value, and permissions to define if the attribute can be read or written [43]. GATT is responsible for defining how data or attributes are packaged, formatted, and exchanged between connected devices. It establishes a framework for managing data in a BLE device, also handling the data interaction between application and stack layers in connection-based communication, unlike GAP, which focuses on low-level interactions with devices and handling the general link management. The security aspects in BLE communication are being managed through SMP [19], [44].

PHY layer is responsible for transmitting raw bitstreams over PHY medium by defining the electrical and PHY specifications for device connections; it also handles the encoding and decoding of binary data into signals. Meanwhile, the logical layer manages the radio's behavior and optimizes it for specific application traffic types. It also defines the use case of physical channels through a state machine to control the

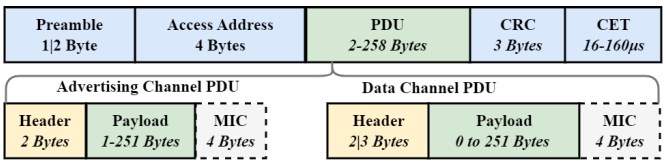


Fig. 2. BLE frame structure based on core specification 6.0.

radio's behavior to suit different data types, e.g., isochronous data. IAL is used to provide segmentation, fragmentation, reassembly, and recombination services for converting service data units (SDUs) from the upper layer to protocol data units (PDUs) of the logical layer and vice versa [44].

Communication between two BLE devices is managed through a link layer using two modes of communication: connectionless communication mode, where one device acts as an advertiser by broadcasting the data packets and the other as a scanner to receive the packets, and connection-based communication mode, where both advertiser and scanner devices are able to establish bidirectional connection and adopt the role of central and peripheral, respectively [19].

The BLE frame structure is divided into two basic formats: the frame structure for basic data rate (BR) and the frame structure for EDR. The basic rate frame structure consists of three parts: access code, header, and data, where the access code is a 9-bytes field containing SYNC bits to identify the controller. The header is a 54-bits field containing 18-bits patterns repeated three times, with subfields including address and type. The address subfield can be identified as high as seven peripherals (1–7), where address 0 renders broadcast. The type subfield identifies the type of data from upper layers, e.g., asynchronous connection-less (ACL), poll, or null. Data is a variable-length field ranging from 0 to 343 bytes; it controls information from upper layers.

EDR frame structure, on the other hand, contains additional fields, e.g., a guard field and a trailer, where the guard field enables switching to higher data speed while transmitting data field and is a 2-bytes field containing a SYNC pattern. A general frame structure of BLE is represented in Fig. 2.

Every logical layer packet contains a preamble that the receiver uses for frequency SYNC and symbol timing estimation, where a predetermined pattern of alternating 0 and 1 bit makes up the preamble. An access address serves to identify a specific connection and help prevent packet collisions on shared radio channels. The access address is a 32-bit field that is unique to each connection, ensuring that packets from different connections do not mix up even if they are transmitted on the same physical channels.

PDU is a unit of information transmitted between peer entities at the same layer of the network. It is classified into two types: advertising channel PDUs and data channel PDUs [44]. The advertising channel PDUs are used for broadcasting data, e.g., BLE Beacons, and they contain different packet types. The data channel PDUs, on the other hand, are used for data transmission in established connections. The cyclic redundancy check (CRC) ensures the integrity of the beacon frame content by validating the frame to see whether it has been tampered with or corrupted during transmission and ensuring that the data received is intact and reliable. The

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concept of constant tone extension (CTE) was introduced in BLE 5.1 to support AoAs and AoDs' DF feature [1].

CTE is a sequence of bits with a variable duration from 16 to 160 ms, which is transmitted at a constant frequency and wavelength in order to enable phase measurement for DF in SAPMs. The signal in the CTE field is not subject to whitening, which is a process that scrambles signals to avoid long strings of 1s or 0s, ensuring a constant tone for accurate phase measurement. Hence, CTE aims to enhance positioning accuracy up to the submeter-level [31].

BLE's stack architecture keeps changing, which facilitates secure communication and accurate positioning estimation through its layered structure and specific frame designs. To address the primary challenge of signal interference with other wireless technologies, e.g., Wi-Fi, which can cause packet loss if they collide with each other on the same frequency channel during transmission, BLE addresses this issue using frequency hopping spread spectrum (FHSS). Specifically, adaptive frequency hopping (AFH), which mitigates interference by identifying a clear transmission path to avoid collisions [45]. This technique divides the frequency range from 2400 to 2480 MHz into different nonoverlapping channels, each with a 2 MHz bandwidth, and rapidly hops between these channels during packet transmission [46]. BLE dynamically adjusts its hopping sequence to minimize interference, tracking, and avoiding busy and noisy channels in real-time.

While BLE has a huge domain of applications, it has evolved significantly as a core enabler of IPS used in the real world, e.g., shopping malls, hospitals, warehouses, museums, and parking lots. The known technologies and platforms that have implemented BLE are Apple iBeacon for proximity solutions, Google Eddystone an Android-based beacon for proximity solutions, Estimote a beacon solution with analytics SDKs, Kontakt.io a location-based infrastructure for healthcare, BlueCats a tracking solution in industrial IoT, and Quuppa a solution for high precision using the DF feature.

B. UWB Technology

UWB is a general radio communication term using a bandwidth close to or greater than 500 MHz. The development of UWB technology started toward the end of the 1990s, and after the Federal Communication Commission (FCC) published the new regulation in 2002, which was then used in both industries as well as academia [47].

In the following year, the WPAN standard was specified by the IEEE 802.15 working group, where 802.15.3 was leveraged for high-rate UWB technology and 802.15.4 standard for low-rate communication applications by including the multiband orthogonal frequency division multiplexing (MB-OFDM) as well as the pulsed direct sequence UWB (DS-UWB), which are still being used, e.g., IEEE 802.15.4z/4 utilizing the Pulsed DS-UWB for IPS and security of applications.

IEEE 802.15.4a was the first UWB standardization released in 2007 that defines the MAC and PHY layers. It was focused on impulse radio UWB (IR-UWB), which uses short, high-bandwidth pulses for communication, offering advantages like low complexity, high data rates, low-power consumption, and fine time resolution. This makes IEEE 802.15.4aa a suitable option for different types of IPS applications [48].

Later in 2011, IEEE 802.15.4a amendment was incorporated into the main body of the standard, after which IEEE released the 802.15.4f amendment in 2012 by adding an additional UWB PHY layer called low rate pulse (LRP). However, the added layer enables UWB to transmit less data, but it carries a stronger pulse from 6.28–9.18 GHz spectrum. The IEEE 802.15.4f amendment was then incorporated into the main body standard of IEEE 802.15.4 in 2015, which specifies two types of UWB PHY modes, the HRP and LRP. The correspondence of HRP is with 2011's (802.15.4) amendment PHY specifications, and the LRP mode corresponds with 2012's (802.15.4f) amendment PHY specification. Energy transmission is the same for both, which is limited by the maximum mean power spectral density (PSD). HRP mode has a high transmission rate with lower pulses, whereas LRP has a low transmission rate with high pulses [49].

In 2020, IEEE introduced the 802.15.4z amendment to the existing 802.15.4 standard, enhancing the UWB PHY with two main objectives: security and positioning accuracy [49], [50]. These improvements include additional coding and preamble options like adding a scrambled timestamp sequence (STS) to security and preamble, and start-of-frame delimiter (SFD) codes for accuracy. This enabled UWB with higher pulse repetition frequency (PRF): 128 and 256 MHz and a new modulation named binary phase shift keying (BPSK) to the HRP mode. In LRP mode, enhancements include support for higher data rates up to 10 Mb/s, higher PRFs reaching 4 MHz, modulation schemes such as pulse-based frequency-shift keying (PBFSK) and PBFSK combined with pulse position modulation (PPM), along with added ranging support [35]. The 802.15.4z standard achieves almost three times higher energy efficiency than the 802.15.4 [20]. A summary of the added features to UWB's standards is represented in Table III.

UWB technology uses two types of modulation schemes: the IR modulation scheme and the frequency modulated UWB (FM-UWB) scheme. The IR modulation scheme is basically designed for high data rates and can achieve a considerably low energy per bit, but it requires precise timing SYNC, making the receiver more complex. It uses narrow pulses that can be modulated using ON–OFF Keying (OOK), PPM, phase-shift keying (PSK), or frequency-shift keying (FSK) modulation. Although only OOK and PSK are supported by the standard. On the other hand, FM-UWB, also known as a low-power, low-complexity modulation scheme targeting low to moderate data rates in different applications, e.g., wireless body area networks (WBANs) [51]. FM-UWB signals are generated by concatenating continuous-phase Gaussian FSK (GFSK) modulation with wideband FM. Both UWB's modulation schemes are part of the IEEE 802.15.6 standard [52].

The focus of UWB technology has recently shifted toward IR-UWB. It uses RF pulses with a short duration of up to nanoseconds, leading to a large bandwidth. This has several benefits.

1) It supports a high channel capacity based on its high bandwidth, making UWB a low transmission

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TABLE III
COMPARISON OF DIFFERENT UWB'S IEEE STANDARDS

Standards	Specifications
IEEE 802.15.4a Released in 2007	UWB PHY Standardisation that contained: • BPM+BPSK Modulation • Support of Precise Ranging • 0.11, 0.85, 6.81 or 27.24 Mbps Data Rates
IEEE 802.15.4f Released in 2012	Addition of LPR in PHY which includes: • No Precision Ranging • $0.0312, 0.250, 1Mbps$ Data Rates • PRF of 1 or $2MHz$ • PPM or OOK Modulation
IEEE 802.15.4 Released in 2015	Incorporation of "f" into main body of standard Addition of LRP and HRP modes in UWB PHY • HRP from 2011 (IEEE 802.15.4) • LRP from 2012 (IEEE 802.15.4f)
IEEE 802.15.4z Released in 2020	Enhancement of UWB PHYs (HRP and LRP) with improved SS - TWR and Simultaneous Ranging • Additional Preamble and SFD codes (HRP) • Higher PRFs: 128 and 256MHz (HRP) • New Modulation: PBSK (HRP) • Enhanced Security : STS (HRP) • Higher Data Rate Up to $10mbps$ (LRP) • Higher PRF: 4MHz (LRP) • PBFSK & PBFSK+PPM Modulation (LRP)

Point: In 2011 the IEEE 802.15.4a was standardized to the main body of IEEE 802.15.4 standard with editorial changes.

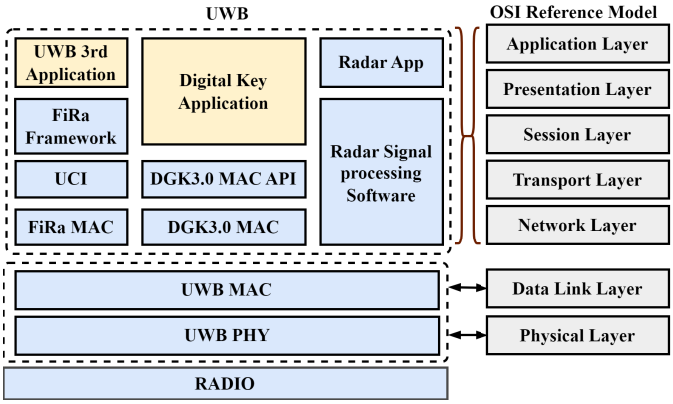


Fig. 3. UWB-based architecture for IoT applications.

power technology, which is useful to avoid narrowband interference.

2) The short duration of UWB impulses reduces the impact of multipath effects, as the pulses arrive at the receiver in a well-separated and filtered manner. This makes UWB inherently robust to multipath propagation, contributing to improved positioning accuracy [53].
3) Due to the rising edge of high temporal resolution being very steep, the timing precision can be enhanced at the receiver side using methods like ToF, TDoA, and TWR [35]. UWB's IEEE 802.15.4 HRP and IEEE 802.15.4z standards architecture is represented in Fig. 3.

UWB's MAC layer is designed based on hardware acceleration (HWA) to ensure low latency and power consumption. The

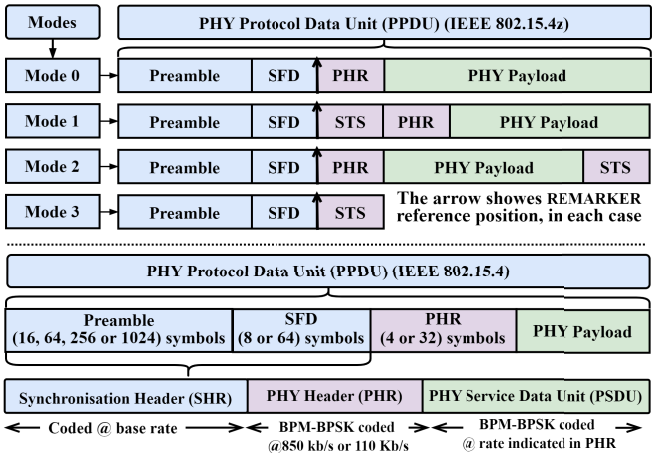


Fig. 4. UWB 802.15.4z standard HRP-ERDEV PPDU formats.

UWB software extensions consist of FiRa-compliant MAC and the car connectivity consortiums (CCC's) Digital Key 3.0 MAC. Furthermore, UWB's architecture supports radar signal processing, as illustrated by applications such as child presence detection (CPD). The software can also be run on general-purpose microcontroller units (MCUs) and can potentially be integrated with BLE as a hybrid system-on-chip (SoC) solution.

According to UWB's standards summarized in Table III, the amendment to the IEEE 802.15.4 standard is the IEEE 802.15.4z. IEEE 802.15.4z defines four different modes of frame structures, each structure has a specific pattern with different objectives [49]. Its frame structures modes (0–3) consist of five different fields, such as preamble, SFD, PHY header (PHR), STS, and PHY payload. Mode 0 frame structure is equivalent to the IEEE 802.15.4 standard frame structure, whereas the other modes (1–3) are different, containing STS fields [35].

In order to establish communication between two different UWB radios, it is important to have the same configured frame structure for both devices. For instance, UWB radios that only support the IEEE 802.15.4 standard may not be able to use the STS field; therefore, they cannot use the other three frame structures, which contain the STS field. The frame structures of UWB's IEEE 802.15.4z standard are illustrated in Fig. 4.

The preamble, also called SYNC, is used to synchronize time and frequency between transmitters and receivers. It contains a sequence of symbols that helps the receiver to align with the incoming signal. The length of the preamble field can vary depending on UWB's configuration and the required SYNC accuracy. SFD marks the beginning of the UWB frame and helps the receiver to identify the start of data transmission, which is a specific pattern that is distinct from the SYNC and other fields. SFD typically has a fixed length, often a few symbols long [54].

PHR contains information about the length of the PHY payload, which contains parameters like length—the number of octets in the PHY payload; rate—the data rate of the PHY payload; the coding scheme used for the PHY payload; and other additional control information, such as error detection and correction schemes. It usually has a fixed length and

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is specified in the standard. PHY payload carries the actual data which is being transmitted; it contains the user's data, with different types of information specific to the type of applications, e.g., sensor data, control commands, or location information. PHY payload has a variable length, which is specified in PHR.

STS, however, is an optional field in frame structure, but a very important field in IPS applications. It is used for fine-grained timing and location services, which help in precise time-of-arrival (ToA) measurements. It works like a preamble but does not repeat itself. STS is a scrambled sequence of symbols that is unique for each transmission. The length of STS can vary depending on UWB's specific configuration and the required timing accuracy. These frame structures (0–3) are designed to enhance the functionality and reliability of UWB communication systems, particularly in scenarios requiring precise timing and location services [35].

Some of the known platform that offers UWB-based services are the Apple U1 chip used in iPhones, Apple Watch, AirTags, AirDrop, and CarKey for tracking. Samsung UWB in SmartThings, Google UWB in Find My Device, and Pixel. Decawave (Qorvo) in industrial applications, NXP in automotive and access control systems, and Sewio RTLS used for high-precision indoor tracking [35].

Unlike BLE, UWB technology has different location device classification (LDC) mitigation in different regions. The modern UWB regulations allow unlicensed operation within the frequency range of 3.1–10.6 GHz, with a low effective isotropic radiated power (EIRP) level, making it suitable to transmit data using pulses instead of continuous waves, which is used in narrowband technologies like BLE.

UWB regulations to operate on UWBs differ from region to region; for instance, Europe, Japan, China, Canada, India, USA, and Singapore each have different unlicensed and LDC mitigation. These LDC keep changing time to time, e.g., before 2015, Canada was using the unlicensed UWB frequency band of 4.75–10.6 GHz. The Canadian Radio-Communications Act (R.S.C. 1985, c R-2) and associated Radio Standards Specification 220 (RSS-220) guidelines were revised in 2015, which expanded the frequency range for UWB operation from 4.75–10.6 to 3.1–10.6 GHz range [52].

Countries like South Korea, Singapore, Japan, and India have LDC mitigation. It is a regulatory mechanism that addresses potential interference from UWB devices, particularly those used for LBS, e.g., different IPS applications, in environments where UWB devices and other RF systems share the spectrum. LDC mitigation contains several techniques, e.g., power control (PC), dynamic spectrum access (DSA), and detection and avoidance (DAA). PC is used to regulate transmission power in wireless systems for maintaining the quality of the signal and reducing cross-channel interference. DSA allows unlicensed users to opportunistically access underutilized licensed spectrum without disrupting primary users. It improves spectrum efficiency in congested environments. DAA is mainly used to ensure safe navigation by adjusting its transmission parameters to avoid causing interference [47].

The choice between BLE and UWB for an IPS application ultimately depends on the specific requirements of an IPS application. This includes its desired accuracy, communication range, budget, and ease of deployment. Some relevant features shared between BLE and UWB through which they can be easily distinguished are given in Table IV.

TABLE IV
BLE'S AND UWB'S FEATURES AND CHARACTERISTICS

Features	Bluetooth Low Energy (BLE)	Ultra-wide Band (UWB)
Standards	IEEE 802.15.1	IEEE 802.15.4/4a/4z)
Coverage Range	Up to $300m$	Up to $200m$
Frequency	2.400 to 2.480 GHz	3.1 GHz to 10.6 GHz
Bandwidth	2 MHz per channel	$>$ 500 MHz
Data Rate	Up to 2 Mbps	Up to 1 Gbps
Modulation	GFSK	BPM and BPSK
Accuracy	$20cm - 1m$	$10 - 30cm$
Security	Secured	Secured
Implementation Cost	Low	Moderate
Positioning Methods	RSSI, AoA, AoD, PBR, RTT	TWR, TDoA, ToF
Transmission Power	0 dBm to 20 dBm	0.5mW to -41.3 dBm/MHz
Power Consumption	Low	Moderate

IV. PRINCIPLES OF POSITIONING IN BLE AND UWB

Location awareness is important to make things more intelligent and offer efficient context-aware service [4]. They eventually require different approaches based on the demand and needs of the IPS application, which holds immense promise with advancements in AI and ML. It is anticipated that they play a pivotal role in overcoming the current limitations and enhancing the capabilities of IPS systems. However, ML algorithms can improve the accuracy of positioning by analyzing large datasets and adapting to changing environments. In addition, the integration of behavioral analytics and contextual awareness will make IPS more intuitive and user-friendly. However, it depends on IPS technologies and their core positioning methods, generating signals and data through which the AI models can be fed.

IPSs are composed of two different components: reference nodes, which have known locations, e.g., satellites in GPS, and Mobile nodes whose location is to be identified. The reference node is also known as an anchor, beacon, and locator [6]. It plays a pivotal role in identifying the position of a Mobile node in an indoor environment. The reference node in BLE is usually called a Beacon, and in UWB, it is called the Anchor node. The role of reference and mobile nodes is specific to BLE's and UWB's positioning methods, especially in terms of transmitting and receiving signals, which is also known as initiator (transmitter) and reflector (receives and transmits), meaning that the reflector, after receiving the signal from the initiator will transmit or respond back to the initiator node. The importance of each component is illustrated in Fig. 5.

With reference to Fig. 5, the position of reference nodes is fixed, having a known location, whereas the mobile nodes (smartphone, vehicle, and tag) have unknown positions; hence, using reference nodes to identify their positions. For the sake of simplicity, this article uses unified terms for the reference node and mobile node. The reference, beacon, anchor, reflector, and locator nodes are referred to as Rn, and the mobile node is classified into two types based on the nature of its communication. The mobile node that either receives or transmits the signal from or to Rn is denoted as Mn, e.g., in

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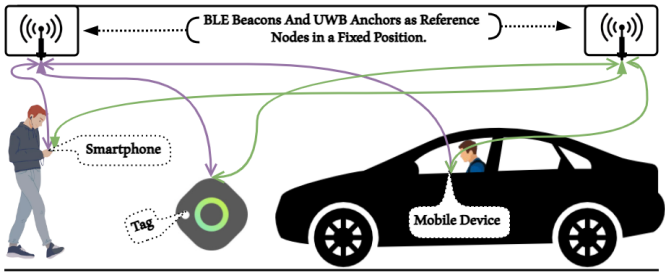


Fig. 5. Indoor-based reference node and mobile node.

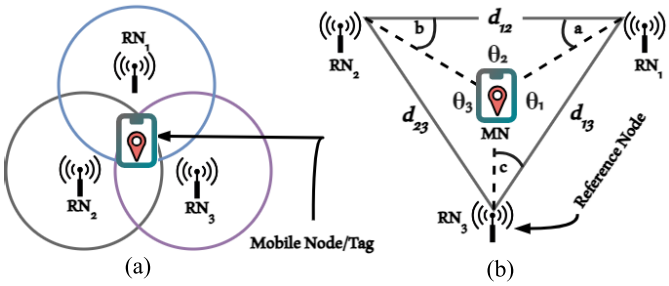


Fig. 6. Trilateration and triangulation techniques. (a) Trilateration. (b) Triangulation.

AoA. On the other hand, a mobile node which transmits the signal to Rn and receives it back is denoted as iMn, which represents the initiator mobile node, e.g., in PBR. The iMn notation is specifically used in positioning methods having a bidirectional communication nature, e.g., RTT, PBR, and ToF, unless mentioned otherwise, whereas notation Mn, apart from being used in SSPMs and SAPMs, it has also been used for general purposes, where a mobile node is referred to.

The most known approaches to identify the position of a Mn utilizing BLE and UWB technologies are trilateration and triangulation. Trilateration approach uses distances from an unknown Mn point to three or more known Rn points to calculate its position [4]. On the other hand, the triangulation technique uses angle measurements for DF, unlike trilateration, which only uses distance measurements. Using the triangulation technique, any point in 2-D/3-D, given the angles between Mn and Rns points, can be calculated, where a minimum of two angles is required to estimate the location of any Mn. The triangulation approach is mostly used in SAPMs, whereas the trilateration approach is mostly utilized in SSPMs and STPMs. An architectural diagram illustrating trilateration and triangulation techniques is given in Fig. 6.

Fig. 6(a) represents a trilateration technique where the intersection point of the Rn indicates the location of Mn. And Fig. 6(b) illustrating triangulation technique, the distances between Rn$(1, 2)$, Rn$(2, 3)$, and Rn$(1, 3)$ are represented by the triangulation parameters (d_{12}), (d_{23}), and (d_{13}), respectively. The angles a, b, and c denote the known angle measurements between the Rns and Mn. These measurements allow for the computation of the angles θ_1, θ_2, and θ_3, which in turn help determine the target position [6].

A. Classification of Positioning Methods in BLE and UWB

BLE and UWB are the most reliable technologies in IPS applications due to their positioning precision and low-energy consumption. It is immensely important to understand the nature of their positioning methods in different IPS applications to identify the position of a Mn [4], [21]. Due to the distinct nature of positioning methods in BLE and UWB, this article has classified them into three different categories based on the signal strength, angle, and time.

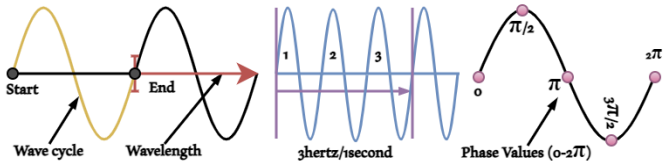


Fig. 7. Wave cycle, wavelength, and frequency.

1) Signal Strength-Based Positioning Methods: SSPMs are used to determine the position of a Mn by measuring the strength of the signal, which is either received or transmitted from a known Rn. Most commonly used SSPMs are RSSI and fingerprint methods. The terms RSS and RSSI are commonly used to indicate the actual power of the transmitter's signal strength at the receiver side. It is basically measured in decibel-milliwatts (dBm) or milliwatts (mW). Despite RSSIs and RSSs' shared focus on signal strength, they differ in their definitions and practical applications [21].

RSS refers to the absolute value of signal strength in a standard unit of measurement, which is dBm (decibels relative to 1 mW). It gives a more precise and measurable value of the received power. In the context of industrial IoT, precise RSS measurements are vital for assessing the communication range of wireless devices, signal penetration capabilities, and link quality. In the RSS approach, the value is always negative, where a number closer to 0, e.g., −20 dBm, indicates a stronger signal and a number far from 0, e.g., −89 represents a weaker signal [3], [55]. The reduction of the measured signal strength as the Mn gets away from the Rn is called attenuation or path loss, such that $p = \text{Txp} - \text{RSS}$, where Txp is the transmission power and p is the path loss.

On the other hand, RSSI is a relative measurement of the signal strength that a device receives from a transmitter, often on a scale with arbitrary units. It is often device-specific and does not have a standardized unit like dBm, e.g., in Wi-Fi systems, it might range from 0 to −100; the closer to 0, the stronger the signals and vice versa. RSSI can be less precise in comparison to RSS due to its simplified or abstracted version of the actual signal strength, which is internally used by hardware to assess how strong the incoming signal is [21] and [55]. Some common terms, such as signal strength, wave cycle, wavelength, and phase value, which will be used throughout this article, are illustrated in Fig. 7.

The amplitude of a radio wave corresponds to the energy it carries and is referred to as the signal strength, which oscillates above and below a central reference value. This oscillation (the above and below) regularly repeats, where a signal transitions to the peak amplitude, down to the trough, and back to its starting point, is called a wave cycle. A signal wave cycle has a physical length, and a signal's wavelength is related to the frequency, where frequency is the number of complete wave cycles that pass over a fixed point in space in one second. The

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wavelength of a signal is measured in hertz (Hz), where 1 Hz represents one wave cycle per second. In consideration of the wave cycle, a phase is a point that sits within a single wave cycle, which is expressed by an angular measurement, and the phase value has 0°–360° range or 0–2π radians [56].

Frequency and wavelength are inversely related to each other; the higher the frequency, the shorter the wavelength, and vice versa. There are some mathematical equations that can explain these concepts easily, e.g., $\lambda = (c/f)$, which illustrates the calculation of an unknown wavelength (λ) from a known frequency (f) and a constant speed of light (c), which is approximately 3×10^8 m/s. An unknown frequency f from a known wavelength λ in a constant c can be computed as $f = (c/\lambda)$ and $c = f * \lambda$, which is the speed of light using a frequency f and a corresponding wavelength λ.

2) Signal Angle-Based Positioning Methods: SAPMs are based on angle measurements between Rn and Mn to determine the position of Mn [23]. SAPMs can provide enhanced positioning accuracy, as they can be implemented using narrowband antenna arrays. Although designing a complex antenna can be costly, which makes the IPS application expensive to implement [3]. SAPM's methods are useful in scenarios where direct LoS communication is not possible due to obstacles or the complicated structure of a building. The most commonly used SAPMs are the AoA and AoD methods. BLE and UWB technologies both support SAPMs; however, BLE is mostly known for it.

3) Signal Time-Based Positioning Methods: STPMs rely on measuring the time it takes for signals to travel between Rn and Mn for measuring the distance between them. STPMs have a broad implementation in UWB technology, specifically the ToF and TDoA methods. They provide a higher range of positioning accuracy in UWB compared with BLE-based methods [15]. BLE technology has acquired STPM as part of its new release of BLE 6.0 by introducing the concept of CS in 2024. The CS in BLE consists of two positioning methods, such as RTT and PBR methods, which offer enhanced security mechanisms in BLE [41].

Ranging techniques in STPMs can be categorized into two schemes: the synchronous ranging and the asynchronous ranging schemes. The synchronous ranging schemes require precise clock SYNC between Rn and Mn for accurate measurement of the signal's arrival time or its TDoA, e.g., the TDoA method, ensuring high positioning accuracy. On the other hand, asynchronous ranging schemes do not require strict clock SYNC; instead, they use those methods that estimate and compensate for clock offsets, e.g., RTT method or least-squares estimators, allowing for accurate positioning without the need for synchronized clocks [57].

Although TDoA is theoretically available in BLE technology, unfortunately, it is confronted with SYNC difficulty due to cross-SYNC between the BLE devices required for precise time measurements [4], [41].

SSPM, SAPM, and STPM are furthermore subbranched into different positioning methods, where each has a distinct nature of communication offering various types of services to different types of IPS applications. The BLE-based positioning method evaluated in this article covers all the categories, while UWB is an STPM-focused IPS technology, although it supports the SSPMs and SAPMs. Fig. 8 represents the classification of BLE's and UWB's positioning methods.

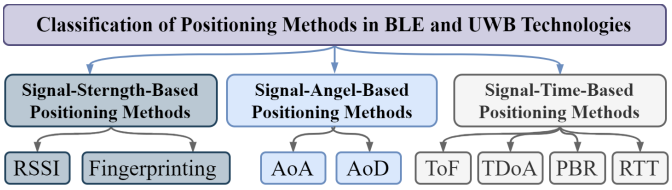


Fig. 8. UWB- and BLE-based positioning methods and their classifications.

B. BLE-Based Positioning Methods

The implication of different positioning methods, e.g., SSPM using RSSI and Fingerprinting, SAPM using AoA and AoD, and STPM using PBR and RTT in BLE, is comparatively larger than UWB in terms of their applications. This is due to the demand for their implementation simplicity and low cost of BLE technology, which are the primary concerns of IPS applications. We have evaluated each method based on its positioning accuracy, implementation simplicity, and resource utilization in an indoor environment.

1) Received Signal Strength Indicator: RSSI is an SSPM, which is widely used in different IPS applications due to its implementation simplicity and has low hardware requirements compared with other positioning methods [55], [58]. It uses the trilateration approach for identifying the positioning of a Mn. The estimated positioning accuracy in the RSSI method depends on the strength of the signal and its propagation model [4], [21]. In BLE, RSSI is reported as part of the advertising packets and connection parameters to estimate distance based on signal strength [59]. In order to identify RSSI's values, one possible approach is to use the log-distance path-loss model, which can be expressed as

$$\text{RSSI} = -10_n \log_{10}\left(\frac{d}{d_0}\right) + A \tag{1}$$

where n is the path-loss exponent, d is the distance between Rn and Mn, d_0 is the reference distance (usually 1 m), and A is the RSSI's value at the reference distance. The path-loss exponent n varies depending on the environment and can range between 2 and 4. Log-distance path-loss model predicts the path loss a signal encounters inside a building or densely populated areas over long distances, which is expressed as

$$L = L_{\text{Rn}} - L_{\text{Mn}} = L_0 + 10_\gamma 10_{10}\left(\frac{d}{d_0}\right) + X_g \tag{2}$$

where γ is the path-loss exponent, d_0 is the reference distance, and X_g is a normal (Gaussian) random variable reflecting the attenuation caused by flat fading. In practical cases, the path-loss exponent n is determined based on the environment and can be influenced by factors such as the presence of obstacles and environmental materials. Hence, the distance between Rn and Mn can be computed as

$$d = d_0 \cdot 10^{\frac{A-\text{RSSI}}{10n}}. \tag{3}$$

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To determine the position of a Mn in reference to Rns having known positions. The distances from at least three Rn should be utilized for applying the trilateration technique, so that the positioning of the Mn will be identified. Let us assume that there are three Rns located at known points (x_1, y_1), (x_2, y_2), and (x_3, y_3) with corresponding distances d_1, d_2, and d_3. Thus, to describe the relationship between the (x_1, y_1), (x_2, y_2), and (x_3, y_3) in order to identify any point on the circle based on the trilateration approach. We can identify a target position in a given circle by representing it in a 2-D space using nonlinear least squares equations for every pair of the Rn and Mn in our case three Rns, such that, $(x - x_1)^2 + (y - y_1)^2 = d_1^2$, $(x - x_2)^2 + (y - y_2)^2 = d_2^2$, and $(x - x_3)^2 + (y - y_3)^2 = d_3^2$, where (x, y) represents the Mn position with respect to the distances d from Rn. In consideration of the given notations, we can find the coordinates (x, y) of all the three pairs by subtracting them from each other, e.g., $(x - x_2)^2 - (x - x_1) + (y - y_2)^2 - (y - y_1) = d_2^2 - d_1^2$, which results in a linear equation based on x and y. This procedure can be followed up to find the values for x_2, y_2 and x_3, y_3 [60].

In real-world applications, the communication of Rns with Mn depends on their coverage range, where we might have a scenario in which more than three Rns communicate with a Mn; in such a case, we call it a multilateration technique. Let us assume there are n number Rns located at known coordinates, e.g., $\text{Rn}_i(x_i, y_i)$. A nonlinear least squares equation to identify the distance between Mn and ith Rn can be expressed as

$$(x - x_i)^2 + (y - y_i)^2 = d_i^2 \quad \forall i \in \{1, 2, 3, \ldots, n\}. \tag{4}$$

Furthermore, we can use the objective function E for adjusting its parameters to minimize the distance effectively in identifying the Mn's position

$$E = \sum_{i=1}^{n-1} \left((x - x_i)^2 + (y - y_i)^2 - d_i^2 \right). \tag{5}$$

The RSSI method is highly vulnerable to multipath effects and interference, which leads to high positioning errors due to its average amplitude over the whole signal bandwidth and the accumulated signal over all antennas [21].

The analytical results of the evaluated BLE's and UWB's positioning methods in this article are further corroborated by simulation using MATLAB software. The obtained results are evaluated in 20×20 m dimensions indoor location with 60 Mns located in different positions whose locations are to be identified, in consideration of LoS and NLoS conditions, environmental noise, and other indoor-related obstacles.

While in LoS conditions, every Mn is able to receive the signals from all the Rns without the signals being reflected or bounced off. However, the environmental noise and communication delay have not been neglected. On the other hand, in NLoS conditions, apart from considering related noise and communication delay, the real-world obstacles like walls have also been used, where the signals are being reflected or bounced off. The simulated scenarios for both LoS and NLoS conditions are given in Fig. 9.

The results shown in Fig. 9 consist of LoS and NLoS conditions, where in LoS all the Mns are in communication

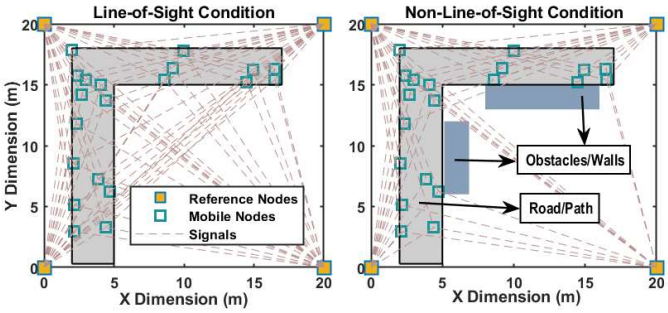


Fig. 9. Simulated environment for LoS and NLoS conditions.

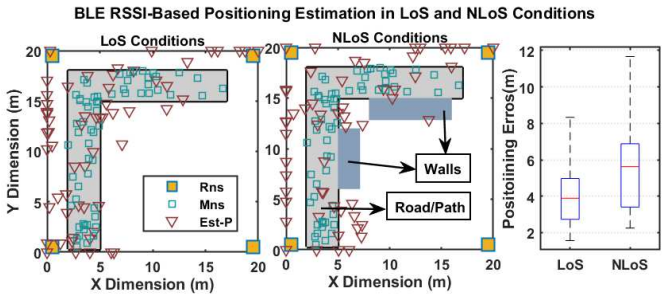


Fig. 10. RSSI results in LoS and NLoS conditions based on 60 Mn.

with the Rns. And in NLoS condition, it can be observed that Rn(0, 0), Rn(0, 20), and Rn(20, 20) could not communicate some of the Mns due to the existence of the obstacle in the environment, which is the wall that bounces off the signals. This eventually impacts the positioning accuracy due to a limited number of communications with Rns, as the trilateration technique requires at least three Rns to communicate with Mn in order to identify its precise position. In the case of NLoS, this is lacking, which leads to additional positioning errors.

SSPMs in NLoS are posed to signal attenuations, which reduces the strength of the signals, leading to disproportionately large positioning errors compared with LoS conditions, e.g., the obtained results of RSSI given in Fig. 10.

The results obtained from the RSSI method in Fig. 10 represent positioning errors for both conditions, where in the LoS condition, having no obstacles, the positioning error is up to 7.5 m. And in NLoS conditions, due to the presence of walls, it goes up to 12 m.

Modern IPS application rarely uses the RSSI method due to its low positioning accuracy; however, some of the applications use the fingerprint-based method, whose principle is based on RSSI, which provides relatively enhanced positioning accuracy than RSSI. This method consists of two stages. In the first stage, we need to collect RSSI values from the environment where the system will be implemented manually and store the collected RSSI values in a database. In stage two, the device collects its current signal readings and matches them with the closest fingerprint in the database. We can use statistical methods or ML algorithms to estimate the position of Mn based on the best match, which can result in better positioning accuracy [61].

2) Angle of Arrival: The release of BLE 5.1 in 2019 has introduced the DF feature as part of its core specification, supporting the SAPMs, e.g., AoA and AoD methods. It has

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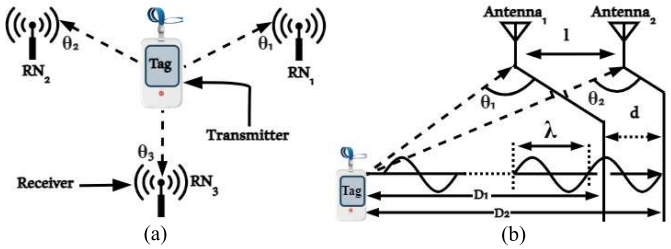


Fig. 11. AoA method's architecture diagram. (a) Trilateration. (b) Triangulation.

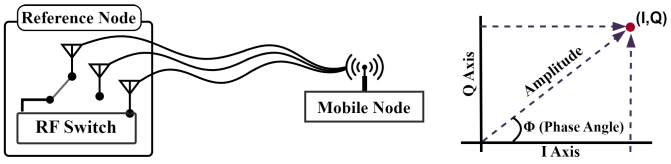


Fig. 12. IQ sample switching signal.

revolutionized the future of Bluetooth technology, which has contributed to a large number of IPS applications since then. The main strength of the DF feature in BLE 5.1 is using an antenna array. This enhances BLE's positioning accuracy with precision up to 2 m in LoS conditions. The importance of antenna arrays is based on their ability to beamform, which guides the signal to a specific direction, enabling the signal to travel for a long distance [62].

The AoA method utilizes the antenna arrays on the Rn side, which act as a receiver, and based on the BLE 5.1 core specification, the Mn side, which is mostly a tag, mainly uses a single antenna for transmitting the signal to Rns [1].

The calculation of the arrived signal angle in Rn in most cases will be performed in a third layer of the IPS application to identify the position of a Mn based on the preidentified locations of Rns. AoA is using a triangulation approach to estimate the position of a Mn as represented in Fig. 6(b). In order to identify the direction of a signal, it is essential to differentiate the phase difference of a specific type of antenna array and the number of antennas used in a Rn. A general diagram representing the AoA method's architecture and the phase difference between antenna arrays is given in Fig. 11.

As mentioned, each Rn is equipped with antenna arrays responsible for receiving the transmitted signal from a Mn. It is then used to identify the position of a Mn. Fig. 11 consists of two parts, where Fig. 11(a), represents an angle-based triangular approach to identify the position of a Mn based on three Rns, e.g., RN_1, RN_2, and RN_3, each with arrival angles θ_1, θ_2, and θ_3, respectively, where these angles determine the direction of arrived signals. On the other hand, Fig. 11(b) illustrates the phase differences between multiple antenna in Rn, in our case two antennas $Antenna_1$ and $Antenna_2$.

The Rn can be equipped with any number of antenna arrays depending on the requirements and complexity of an IPS application. Antenna arrays are commonly classified into three types: uniform circular array (UCA), uniform rectangular array (URA), and uniform linear array (ULA), which have different implications specific to an indoor environment [63].

Phase difference is an important factor in SAPMs, which determines the direction from which a signal is received. The arrival signal's angle in Rn is mainly determined using phase difference by considering the spacing between the antennas and the wavelength of the signals. In accordance with Fig. 11(b), the phase difference between signals arriving at the two antennas spaced at a known distance d of ULA type, where the signal travels in a LoS condition, can be expressed as

$$\Delta\phi = \frac{2\pi d \sin(\theta)}{\lambda} \tag{6}$$

where $\Delta\phi$ (or ϕ) is the phase difference between the signals arrived at two antennas, d is the distance between two antennas in the array, θ represents the AoA, which is the angle between the incoming signal and receiver array's axis, and λ is the wavelength of the signal that is related to the frequency f of the signal and the speed of light c. Phase measurements can be more precise in UWB due to its wide frequency range, e.g., 3.1–10.6 GHz, which has high-frequency signals with short wavelengths, than in BLE, which has a narrower frequency range, typically around 2.4 GHz.

Although the given equation for phase difference is effective in LoS conditions, there is no single closed-form equation for it in NLoS, but the received signal can be modeled as a sum of MPCs as follows:

$$r(t) = \sum_{i=1}^{n} A_i e^{j(2\pi f t + \Delta\phi_i)} \tag{7}$$

where A_i is the amplitude of the ith path, $\Delta\phi_i$ is the phase shift due to path length and reflection, and n is the number of MPCs. Using $\Delta\phi$, in consideration to (6), the arrived angles from a Mn which is θ can be given as

$$\theta = \arcsin\left(\frac{\lambda \cdot \Delta\phi}{2\pi d}\right). \tag{8}$$

The phase of a wave is defined as a specific point in the wave cycle, which can be measured as the wave passes over an antenna. Based on θ, the Mn transmits the DF signal using a single antenna, and the Rn, equipped with antenna arrays, takes the in-phase and quadrature-phase (IQ) samples while switching between the antennas. Rn uses the received IQ samples, which consist of a pair of IQ samples, to calculate the θ, which is AoA [64]. The direction of the signal is determined by varying the propagation delays of the BLE signal between the antenna array and the receiving signal [65]. The process of the RF switch based on the IQ samples is illustrated in Fig. 12.

In the context of IQ samples, the real and imaginary parts of IQ samples represent components of a complex signal that encodes information about the signal's amplitude and phase. A complex signal is composed of a real part I and an imaginary part Q, which is commonly used in wireless communications and signal processing to represent signals that vary over time [64], [66]. IQ samples are derived from the received signal, typically by mixing the received signal with a reference signal (local oscillator) to obtain the IQ components. This allows us

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to capture the amplitude and phase information of the signal [64].

Although SAPMs in BLE and UWB are based on a triangulation approach, however, they can also use a trilateration approach by combining them with STPMs, e.g., ToF. Apart from this, AoA in both BLE and UWB uses a triangulation approach to estimate the position of a Mn in an indoor location.

Using triangulation technique for AoA there are a number of different approaches to identify the position of a Mn, e.g., direction vector, parametric line, and implicit line, in consideration of the θ given in (8), and three Rns as illustrated in Fig. 11, e.g., Rn_1, Rn_2, and Rn_3, at their known coordinates (x_1, y_1), (x_2, y_2), and (x_3, y_3), with arrived angles θ_1, θ_2, and θ_3 transmitted from Mn, respectively.

In this study, we have used the implicit line approach for both AoA and AoD methods, which can be the best approach to identify the intersection points analytically. Thus, the line representing the direction from Rn_1 to the Mn can be expressed as: $y - y_1 = (x - x_1)\tan(\theta_1)$. It can be rearranged as $y = (x - x_1)\tan(\theta_1) + y_1$. The line representing the direction from Rn_2 to the Mn can be expressed as $y - y_2 = (x - x_2)\tan(\theta_2)$ where $y = (x - x_2)\tan(\theta_2) + y_2$. And the line representing the direction from Rn_3 to the Mn can be expressed as $y - y_3 = (x - x_3)\tan(\theta_3)$ which can be rearranged as $y = (x - x_3)\tan(\theta_3) + y_3$.

Based on the obtained direction from Rn_1, Rn_2, and Rn_3, the intersection of these lines will give us the position of a Mn, which is x, y by pairing the Rn_1, Rn_2, and Rn_3 as $Rn_1 = Rn_2$ such that the two expressions for y can be equal to each other as $(x - x_1)\tan(\theta_1) + y_1 = (x - x_2)\tan(\theta_2) + y_2$. It can be rearranged to isolate x as $(x - x_1)\tan(\theta_1) - (x - x_2)\tan(\theta_2) = y_2 - y_1$, by simplifying the given process we can identify the x and y coordinates which gives us the position of a Mn as $x(\tan(\theta_1) - \tan(\theta_2)) = x_1\tan(\theta_1) - x_2\tan(\theta_2) + y_2 - y_1$, thus, based on Rn_1's and Rn_2's intersected line we can get the x coordinate of Mn as

$$x = \frac{x_1 \tan(\theta_1) - x_2 \tan(\theta_2) + y_2 - y_1}{\tan(\theta_1) - \tan(\theta_2)}. \tag{9}$$

In order to get the coordinate of y we substitute x from (9) into either $y = (x - x_1)\tan(\theta_1) + y_1$ or $y = (x - x_2)\tan(\theta_2) + y_2$ into $y = (x - x_1)\tan(\theta_1) + y_1$ as

$$y = \left(\frac{x_1 \tan(\theta_1) - x_2 \tan(\theta_2) + y_2 - y_1}{\tan(\theta_1) - \tan(\theta_2)} - x_1\right)\tan(\theta_1) + y_1. \tag{10}$$

Finally, we can take the third Rn_3 and can substitute the calculated values of x and y into $y = (x - x_3)\tan(\theta_3) + y_3$. It will eventually give us all the interesting lines from Rn_1, Rn_2, and Rn_3, hence the obtained output will give us the position of a Mn. The given procedure can be furthermore generalized as a uniform equation with reference to any number of Rns to determine the position of a Mn

$$y = (x - x_i)\tan(\theta_i) + y_i. \tag{11}$$

Thus, the objective function E of AoA can be expressed as

$$E = \sum_{i=1}^{n} \left((x - x_i)\tan(\theta_i) + y_i - y\right). \tag{12}$$

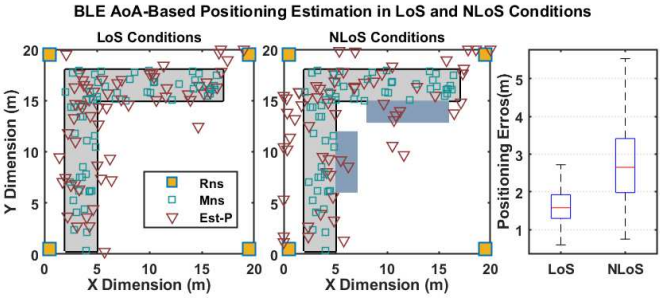


Fig. 13. BLE AoA results in LoS and NLoS conditions.

BLE's SAPMs offer higher accuracy in comparison to its traditional SSPM, e.g., RSSI. This is due to the utilization of antenna arrays that provide strong signal strength in SAPMs. The obtained simulated results in both LoS and NLoS conditions using the AoA method are represented in Fig. 13.

In accordance with the obtained simulated results given in Fig. 13, AoA's estimated positioning error under LoS condition is around 3 m, while in NLoS condition, it can be observed that its estimated error goes up to 5.5 m. As the SSPMs are posed to signal attenuation, on the other hand, SAPMs, apart from being vulnerable to signal attenuation, are also subject to signal reflection. This can increase positioning error in SAPMs by a large margin, as observed in AoA's NLoS condition. A detailed discussion on positioning methods, vulnerabilities in BLE and UWB technologies is carried out in Section V-A.

3) Angle of Departure: The BLE 5.1's other contribution to SAPM is the AoD method. While it shares some common attributes with AoA, e.g., knowing the direction of signals and computing their angles, meanwhile, the main difference between them is the flow of communication and the source of transmitted signals. In AoA, the Mn transmits the signals while Rns receive those transmitted signals, whereas in the AoD method, Mn receives the transmitted signal from Rns. AoD can be considered as one of the most important SAPMs due to its communication flow and architectural design [62].

AoD's communication flow is opposite to AoA, meaning that the Rns in the AoD method transmit the data packet instead of receiving it, while the Mn acts as the receiver capable of computing the departed single or the signal coming from Rns. Its best use case is indoor navigation, which is sometimes referred to as indoor GPS [6], due to its architectural design. The analysis for positioning a Mn can be performed in Mn itself, which can simplify the implementation of the IPS application. The process of AoD's communication between Rns and Mn is illustrated in Fig. 14.

Fig. 14 consists of two parts: Fig. 14(a) represents an angle-based triangular approach to identify the position of a Mn using three Rns, with known location, each with departed angles β_1, β_2, and β_3, respectively, which determines the direction of signals. And Fig. 14(b) describes the phase difference between multiple antennas in Rn, in this case two antennas $Antenna_1$ and $Antenna_2$. To identify the angle and direction of a departed signal, we must first know its phase difference to get its angle. The phase difference in AoD can be

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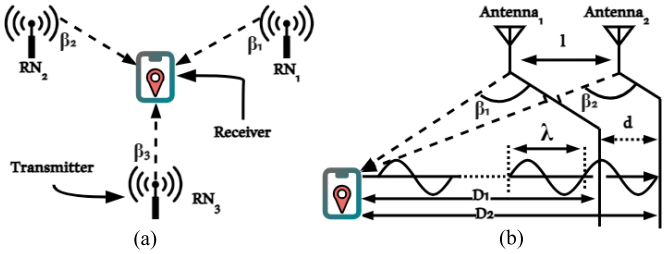


Fig. 14. AoD-based positioning architecture. (a) Triangulation. (b) Phase difference.

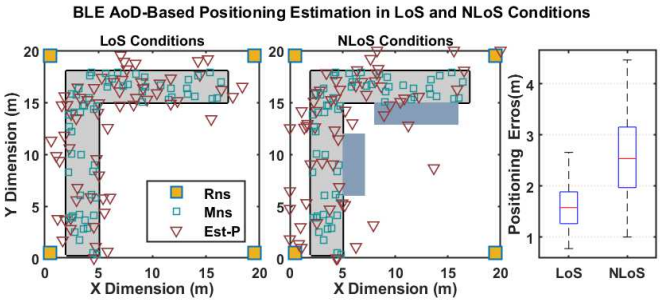


Fig. 15. BLE AoD results in LoS and NLoS Conditions.

expressed as

$$\phi = \frac{2\pi d \sin(\beta)}{\lambda} \tag{13}$$

where ϕ is the phase difference between the signals departed from different antennas, d is the distance between the antennas, and β represents the departure angle of signals from Rn, which is the outgoing signal to be received by Mn. Hence, in reference to (13), β can be computed as

$$\beta = \arcsin\left(\frac{\lambda \cdot \phi}{2\pi d}\right). \tag{14}$$

Although computing the phase difference and angle of a signal, either from Rn or Mn, is the same in BLE's SAPMs. However, due to their different flow of communication, identifying the position of a Mn using the triangulation approach is slightly different in AoD and AoA. In AoD, estimating the position of a Mn can be in Mn itself based on the computed β given in (14). It takes the IQ samples while switching between the antennas, and using the IQ samples to calculate the β, instead of computing β in Rn as was performed in AoA. Hence, it makes the computational process slightly different in triangulation as compared with what was done in the AoA.

Let us assume that the unknown coordinates of Mn are (x, y), which are to be identified based on β. And the three Rns, e.g., Rn_1, Rn_2, and Rn_3, at their known coordinates (x_1, y_1), (x_2, y_2), and (x_3, y_3), each with departed angles β_1, β_2, and β_3, respectively. Thus, the line representing the direction from Rn_1 to Mn can be expressed as: $y - y_1 = (x - x_1)\tan(\beta_1)$. It can be rearranged as $y = y_1 - (x - x_1)\tan(\beta_1)$, the line representing the direction from Rn_2 to the Mn can be expressed as $y - y_2 = (x - x_2)\tan(\beta_2)$ where $y = y_2 - (x - x_2)\tan(\beta_2)$, and the line representing the direction from Rn_3 to the Mn be expressed as $y - y_3 = (x - x_3)\tan(\beta_3)$ which can be rearranged as $y = y_3 - (x - x_3)\tan(\beta_3)$.

Based on the direction from Rn_1, Rn_2, and Rn_3, the intersected point gives us the position of a Mn, which is x, y. By pairing Rn_1, Rn_2, and Rn_3 as $Rn_1 = Rn_2$ such that the two expressions for y can be equal to each other as $y_1 - (x - x_1)\tan(\beta_1) = y_2 - (x - x_2)\tan(\beta_2)$, which can be rearranged to isolate x as $(x - x_2)\tan(\beta_2) - (x - x_1)\tan(\beta_1) = y_2 - y_1$, it can be simplified as $x(\tan(\beta_2) - \tan(\beta_1)) = x_2\tan(\beta_2) - x_1\tan(\beta_1) + y_2 - y_1$, thus, based on Rn_1's and Rn_2's intersected line we can get the x coordinate of Mn as

$$x = \frac{x_2\tan(\beta_2) - x_2\tan(\beta_1) + y_2 - y_1}{\tan(\beta_2) - \tan(\beta_1)}. \tag{15}$$

In order to get the y coordinate, we substitute x from (15) into $y = y_1 - (x - x_1)\tan(\beta_1)$ as

$$y = \left(\frac{x_2\tan(\beta_2) - x_1\tan(\beta_1) + y_2 - y_1}{\tan(\beta_2) - \tan(\beta_1)} - x_1\right)\tan(\beta_1) + y_1. \tag{16}$$

Now we take Rn_3 and substitute the calculated values of x and y into $y = y_3 - (x - x_3)\tan(\beta_3)$. This will eventually give us all the interesting lines from Rn_1, Rn_2, and Rn_3, estimating the position of a Mn. The given process can be generalized as a uniform equation to compute any number of Rns communicating Mn to identify its position as

$$y = y_i - (x - x_i)\tan(\beta_i). \tag{17}$$

Thus, the objective function E for AoD can be defined as

$$E = \sum_{i=1}^{n} (y_i - y - (x - x_i)\tan(\beta_i)). \tag{18}$$

The AoD method is a reliable approach for navigation-based applications. It can achieve higher positioning accuracy compared with the AoA. This is in scenarios where the computation of the departed signals is performed in Mn, which can provide better communication range, hence providing better positioning accuracy. In most cases, AoD-based positioning systems can illuminate the third layer of communication to identify the position of a Mn. In such a scenario, the Mn itself can act as a positioning engine. The simulated results for AoD in both the conditions (LoS and NLoS) are given in Fig. 15.

The obtained simulated results of the AoD method given in Fig. 15 represent positioning errors under LoS and NLoS both. It's estimated error under LoS is 3 m, while in NLoS conditions, it goes up to 5 m. This is due to the reflection of the signals, which leads to extra positioning error in AoD under NLoS conditions. SAPM's estimated errors can be further reduced by using proper antenna arrays or integrating an optimization algorithm with them.

The principles of SAPMs are the same in both UWB and BLE technologies, but their processing methods and PHYs are different. AoD and AoA are most frequently used in BLE-based positioning services. BLE technology got much of the attention in IPS applications due to its SAPMs in 2019, but based on their positioning accuracy and security issues, they were not used in applications requiring centimeter-level

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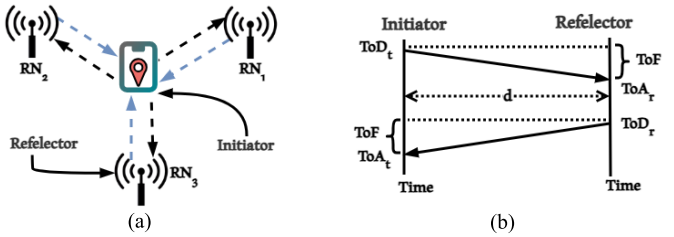


Fig. 16. BLE-based RTT positioning architecture. (a) Trilateration. (b) RTT.

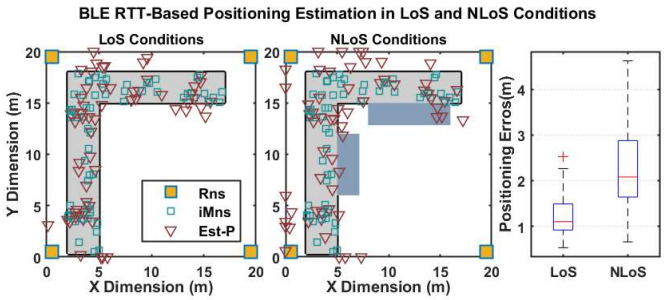


Fig. 17. BLE RTT results in LoS and NLoS conditions.

precision. Hence, Bluetooth technology took another initiative in 2024 by introducing the STPMs such as RTT and PBR.

4) Round-Trip Time: RTT is a BLE-based STPM introduced as part of BLE's CS approach in 2024, which is referred to as the duration a signal takes to travel from iMn to Rn (reflector) and back to iMn. It is a precise ranging method designed to measure the distance between two BLE nodes (iMn and Rn), where the iMn measures the time it takes for the packet to travel to Rn and back, known as RTT. This time measurement is then converted into a distance using the speed of light c to measure the distance. iMn in the BLE core specification is referred to as initiator and Rn as reflector.

RTT is particularly effective in providing secure communication between BLE devices. It can detect and mitigate man-in-the-middle (MITM) relay attacks by verifying the physical proximity of devices. Determining the distance between two BLE devices in RTT is complex than BLE's SAPMs and SSPMs due to the RTT and tracking their timestamp, e.g., iMn sends a poll packet to Rn, which includes a sequence number and a timestamp indicating when the packet was sent. Rn receives the poll packet and immediately responds with an ACK packet, which includes its own timestamp indicating when the packet was received. These two timestamps are being recorded in the iMn, which is the time it sends the poll packet and the time it receives the ACK packet.

Let us assume iMn's timestamp as ToD_i, which is iMn's departed time, and ToA_i, which is the ToA in iMn. On the other side, Rn records the time it receives the poll packet and the time it sends the ACK packet, which is represented as ToA_r, where ToA is the arrival time in Rn and ToD_r is the departed time of Rn. The given process is illustrated in Fig. 16(b), while Fig. 16(a) represents a trilateral approach to identify iMn's position.

By evaluating the ToF based on Fig. 16, which is the time it takes for a packet to be exchanged between iMn and Rn, we can estimate the distance between them. The main aspect here is that both nodes record ToA and ToD. The differences between ToA and ToD can be analyzed in both iMn and Rn, which provides important data for a reliable distance calculation. In consideration of the given points, the RTT between iMn and Rn can be computed as

$$\text{RTT} = (\text{ToA}_i - \text{ToD}_i) - (\text{ToD}_r - \text{ToA}_r). \tag{19}$$

Based on Fig. 16, the RTT in (19) can be rearranged considering the two ToF one from the Rn side and the other from iMn, as RTT = 2 ToF, which accounts for the processing time at Rn, ensuring only the actual propagation time measurement. Hence, the distance between iMn and Rn can be computed in consideration of ToF, the nodes' RTT and speed of light as

$$d = \frac{\text{RTT} \times c}{2}. \tag{20}$$

Although BLE's RTT method has not been fully utilized in applications using multiple Rn to determine the position of an iMn in an indoor environment. However, we have used the trilateration approach for identifying the position of iMn. Defining the role of iMn and Rn in an IPS application is very crucial in CS methods. It can define the nodes' responsibilities to identify their positions using an antenna array.

As of now, BLE's STPMs support up to a maximum of four antennas whose design depends on the type of IPS application and the role of iMn and Rn [41]. In this article, we have balanced the communication between iMn and Rn by equally distributing the antennas between them while considering their nature of communication. The process for identifying the position of an iMn in RTT using the trilateration approach is performed in the same way as in RSSI using the nonlinear least square formula, which can be expressed as

$$(x - x_i)^2 + (y - y_i)^2 = d_i^2 \quad \forall i \in \{1, 2, 3, \ldots, n\}. \tag{21}$$

The objective function E of RTT, which minimizes the sum of squared errors between estimated distances and the distances calculated from iMn, can be expressed as

$$E(x, y) = \sum_{i=1}^{n} \left((x - x_i)^2 + (y - y_i)^2 - d_i^2\right). \tag{22}$$

The obtained simulated results of the BLE-based RTT method under LoS and NLoS conditions are given in Fig. 17.

RTT's positioning errors in both LoS and NLoS conditions are better in comparison to BLE's SAPMs. This is due to its RTT calculation. In accordance with the simulated results given in Fig. 17, RTT's LoS results are under 2.3 m, while its estimated errors under NLoS condition go up to 4.5 m. Although the STPMs provide higher positioning accuracy, they are more vulnerable in NLoS environments, e.g., PBR, which is due to the reflection of the signals.

5) Phase-Based Ranging: PBR is BLE's sophisticated STPM with high positioning precision that measures the distance between two nodes by analyzing the phase difference of radio signals between them. It was introduced as part of BLE's 6.0. Its communication flow is the same as RTT, yet with a totally different mechanism, which is based on radio signal

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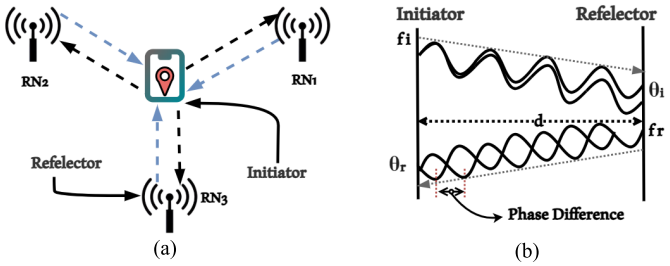


Fig. 18. BLE PBR positioning architecture. (a) Trilateration. (b) PBR.

wavelength [16], [67]. This method is sensible hypothetically in using the phase of a received signal compared with a transmitted iMn's signal to estimate the ToF of the signal. The transmitted signal in space can be determined by calculating the phase difference between the transmitted and received signals. This phase difference is then converted into distance using the known wavelength of the BLE signal, enabling it for precise distance measurements with minimal susceptibility to environmental factors like multipath interference, which is specific to LoS conditions.

PBR operates using a continuous wave (CW) signal, where the frequency of the signal remains constant, enabling phase measurements to be reliable over time. It technically requires SYNC between iMn and Rn, as any phase shift can lead to inaccuracies or errors, especially in NLoS conditions. One of the key features of BLE 6.0 is periodic tone mapping and improved CH, which reduces the impact of interference and enhances the reliability of phase measurements by using the multicarrier phase difference (MCPD) techniques. This enables PBR to illuminate the requirement of a tight SYNC between iMn and Rn.

PBR is particularly effective in mitigating multipath effects, especially when combined with MCPD and CH. It uses tone exchange throughout the 2.4-GHz frequency band, where out of 79 channels, 72 are used for actual transmission of CS tones and packets. The tone exchange in BLE is very effective in mitigating the multipath fading problem by calculating the number of signal phase shifts between iMn and Rn [56]. These two nodes measure phase change over multiple frequencies to identify the distance between them.

The relative distance of iMn will be determined by comparing the phase difference between the signals from both nodes, where the complexity of manipulating phase information makes the PBR more secure [15]. PBR's communication process between iMn and Rn is illustrated in Fig. 18.

In order to estimate the distance between iMn (initiator) and Rn (reflector) in consideration of Fig. 18(b). iMn transmit a signal at a known frequency denoted as f_i whose initial phase is known to iMn. The Rn receives a f_i signal at its antenna by noting its phase as the received phase, which is represented as θ_i. Now Rn echoes the received signal back to iMn with the same frequency, which is denoted as f_r and ensures that the initial phase of the transmitted signal is the same as the received phase of the signal from iMn. After receiving the signal from Rn, iMn will measure the received phase of the signal arriving from Rn, which is denoted as θ_r. The f_i represents multiple frequencies where every iMn's f_i is echoed back to iMn as f_r, each with different phases, e.g., θ_i and θ_r. Now that we have the two phases that include information of phase shift, which was caused by the propagation delay and phase difference between iMn and Rn. Their two-way ranging distance can be computed based on the phase difference for f_i and f_r, which is $\theta_i - \theta_r$.

If the distance d between iMn and Rn is less than the signal's wavelength, which is $(2 \times f/c)$ where f represents the RF ton frequency, and c is the speed of light, then the phase difference or measured phase offset θ can be expressed as

$$\theta = 2\pi d \times \frac{2 \times f}{c}. \tag{23}$$

In a real-world application scenario, the measured distance d must be longer than the signal's wavelength $d > (2f/c)$, in which the number of whole cycles that have elapsed when the RF tone was propagating between two nodes needs to be tracked [31]. Let n be the number of whole cycles (integer) that have elapsed; hence, the distance d can be measured as

$$d = \frac{c}{2f}\left(\frac{\theta}{2\pi} + n\right). \tag{24}$$

Now, we need to track the total number of cycles elapsed using multicarrier phase ranging. To eliminate the tracking of the entire number of cycles elapsed, where the phase measurements are the difference in the phase of the received signal and its own local oscillator signal, taken at multiple RF tone frequencies. The RF signals at different frequencies travel the same distance with the same propagation time, and in return, we can have a different phase shift of offset [16], [31]. Let f_1 and f_2 be two frequencies at a distance d between iMn and Rn. These nodes first perform PBR at RF signal f_1 and then at RF signal f_2, the phase differences measured at f_1 and f_2, can be then calculated for both the θ_1 and θ_2 as

$$\theta_1 = 2\pi\left(d\frac{2f_1}{c} + n\right), \quad \text{and} \quad \theta_2 = 2\pi\left(d\frac{2f_2}{c} + n\right). \tag{25}$$

Hence, the phase differences θ_1 and θ_2 from (25) can be combined to remove the number of whole cycles (n) ambiguity, as represented in the following equation:

$$d = \frac{c}{4\pi} \times \frac{\theta_1 - \theta_2}{f_1 - f_2}. \tag{26}$$

In order to measure phase offset at more than two frequencies to get a better ranging accuracy. We need to consider the slope, which is represented in (27). However, when phase offset measurements are shown as a phase versus frequency curve, the distance d between the iMn and the Rn can then be represented by the slope of the curve as illustrated in the following equation:

$$\theta_i = \frac{4\pi}{c} \times f_i \times d + N, \quad \text{where} \quad d = \frac{c}{4\pi}\text{slope}. \tag{27}$$

In the utilization of PBR for identifying the position of iMn, we have used the trilateration approach in the same way as RSSI, using the Euclidean formula, which can be expressed as

$$(x - x_i)^2 + (y - y_i)^2 = d_i^2 \quad \forall i \in \{1, 2, 3, \ldots, n\}. \tag{28}$$

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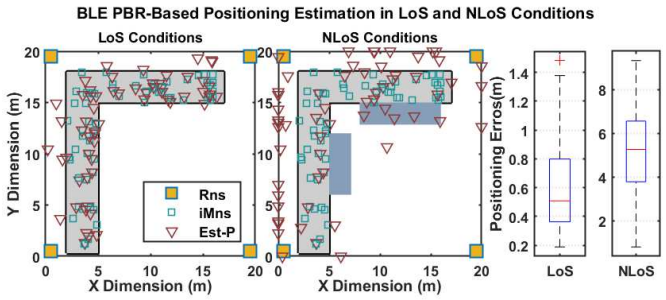


Fig. 19. BLE PBR positioning results in LoS and NLoS conditions.

The objective function E using PBR to minimize the sum of squared errors between estimated distances and the distances calculated from the hypothesized position of iMn, is given as

$$E(x,y) = \sum_{i=1}^{n} \left((x - x_i)^2 + (y - y_i)^2 - d_i^2\right). \tag{29}$$

Although BLE's PBR method achieves higher positioning accuracy in comparison to BLE's other positioning methods, due so the measurement of the phase difference of signals across multiple frequencies in LoS conditions. However, its accuracy degrades by a large margin while dealing with NLoS conditions if it is used without the utilization of MCPD. The obtained simulated result of the PBR is represented in Fig. 19.

It can be observed that PBR is vulnerable to NLoS conditions. This is due to the phase information used for distance estimation and additional phase shifts, which can be significantly affected by obstacles leading to multipath propagation, signal reflections, and distortions that can cause errors in the phase measurements. This results in inaccurate distance calculations, as phase differences are no longer reliable indicators of the true distance between nodes as observed in Fig. 19. However, its accuracy can be enhanced by using MCPD or any other optimization algorithm. On the other hand, PBR's positioning accuracy in LoS conditions is remarkable, which is under 1.5 m.

C. UWB-Based Positioning Methods

UWB technology is mainly focused on STPMs due to its ability to achieve high positioning accuracy by measuring the time signals take to travel between Rn and Mn, e.g., UWB's ToF and TDoA methods.

In addition, high data rate and low latency make UWB technology a suitable choice for real-time applications. Although not all STPMs provide accurate positioning estimation in NLoS conditions, as witnessed in BLE's PBR method. On the other hand, however, UWB, due to its unique characteristics and advanced error mitigation techniques, and large bandwidth, allows for capturing precise ToF measurements of signals. This makes UWB less affected by multipath propagation in NLOS conditions compared with BLE. In contrast, BLE relies on shorter range, lower bandwidth signals, which are more susceptible to interference and signal degradation in NLOS conditions.

The choice of UWB positioning method depends on the specific requirements of an IPS application. It includes the desired accuracy, system complexity, battery life, and cost. While STPMs are ideal for applications demanding high precision, on the other hand, SSPMs and SAPMs are suitable for applications that require moderate precision and are computationally less expensive in comparison to STPMs. For instance, AoA and AoD methods typically rely on a multiple antenna array, where the angle of a signal is determined by using the phase difference between the signals received by the antennas (AoA) or transmitted by antennas (AoD) [68]. In this article, we have mainly focused on UWB's STPMs, e.g., ToF and TDoA, due to their high rate of use cases in different types of IPS applications.

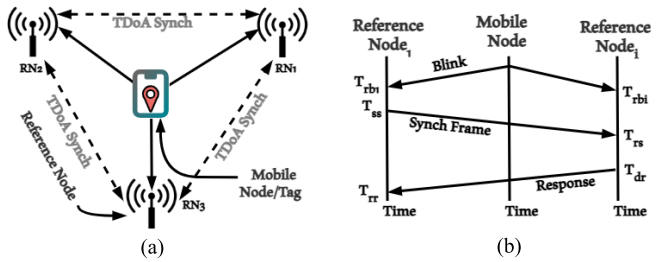


Fig. 20. TDoA architecture and SYNC process. (a) Trilateration.(b) TDoA.

1) Time Difference of Arrival: TDoA is a UWB-based synchronous ranging scheme method, where Mn periodically broadcasts the signal, which contains its departure time. The Rn receives and measures the arrival time of the signal. It involves measuring the time difference between the signals arriving at different Rns to determine the position of a Mn. The Mn sends out a signal, and all nearby Rns receive it and record its arrival time. Since the Rns could be at different distances from a Mn, the signal reaches each Rn at a different time. These time differences, multiplied by the speed of light c, give the distances between Mn and each Rn, which are then used to calculate Mn's position using a trilateration approach.

TDoA's efficiency relies on the Rn's accurate hronization to run on the same clock. hronization is very crucial in the TDoA method; any discrepancies can lead to significant positioning errors [69]. A one-way or unidirectional ranging procedure can be easily implemented due to hronization between the nodes globally. However, every node's hronization is not an easy task; it requires a precise and expensive oscillator to mitigate clock drift and clock offset. Clock drift refers to the gradual deviation of a clock's frequency from a reference clock, and clock offset or skew is the difference in time between two clocks at a specific moment [30].

One of the proposed approaches for the TDoA SYNC problem is the SYNC-free TDoA positioning algorithm, which enables the TDoA method to mitigate the impact of clock drift and improve system performance [69], [70]. The communication flow between Rn and Mn in the TDoA method is illustrated in Fig. 20.

Clock SYNC between Rns plays a crucial role in TDoA, which can be very challenging in wireless communication [3]. The Rns start SYNC with each other after receiving a blink frame from the Mn side. With reference to Fig. 20(b), Rn_1 initializes the process in the initial stage; every Rns is running on an independent clock with a different starting time. By

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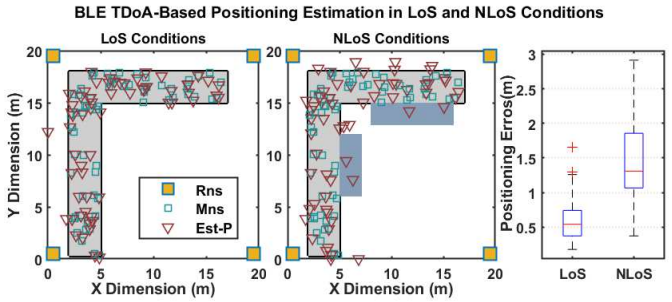


Fig. 21. UWB TDoA positioning results in LoS and NLoS conditions.

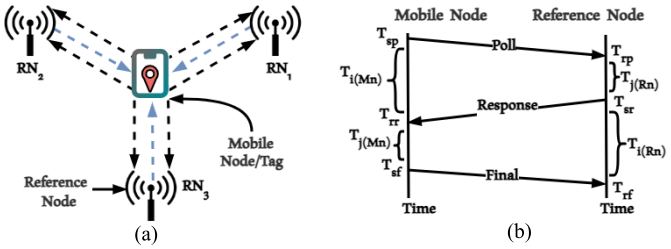


Fig. 22. ToF double-sided two-way ranging architecture. (a) Trilateration. (b) ToF.

finding the clock offsets for all Rns relative to the clock of the controller Rn, which is Rn_1, the SYNC will be achieved, where Rn_1 starts synchronizing the other Rns consecutively by the arrival of a blink frame at time T_{rb} and sends SYNC to the other Rns. Meanwhile, it records the transmission timestamp of the sent frame, denoted as T_{ss}.

The leftover Rns, in this case Rn_i where $\forall i \in \{2, 3, \ldots, n\}$, receive the frame and note the SYNC arrival time T_{ss}, and initiate a delayed reply at time T_{dr}. The reply frame sent by Rn_i includes three timestamps in its payload section, which are T_{rs}, T_{dr}, and T_{rb}, respectively, representing the received blink time at Rn_i [12]. To determine the position of a Mn using TDoA, the time measurements and distance calculation parameters must be considered. Let T_i be the ToA at Rn_i where $\forall i \in \{1, 2, 3, \ldots, n\}$, hence, the time difference between first Rn and the leftovers Rns can be expressed as

$$\mathrm{DT}_{ij} = T_j - T_i \tag{30}$$

where $i \in \{1, 2, 3, \ldots, n\}$ and $j = 1$ indicating the first Rn. Rns known coordinates are denoted as $\mathrm{Rn}_i(x_i, y_i)$ for n number of Rns. Its hyperbolic equation can be expressed such that for any two pairs of Rns, Rn_j, and Rn_i, where Rn_j represents the first Rn with respect to leftover Rns. To determine the position of a Mn we can simplify its distance equation by defining xj, xi and yj, yi as $xj = x - x_j$, $yj = y - y_j$ and $xi = x - x_i$, $yi = y - y_i$. Thus, taking their time difference, the position of a Mn can be computed as

$$D_{ij} = \left((xj)^2 + (yj)^2\right)^2 - \left((xi)^2 + (yi)^2\right)^2. \tag{31}$$

In a more compact way, if we denote the position of Mn as vector $\mathbf{p} = (x, y)$ and Rn positions as vectors $\mathbf{p}_i = (x_i, y_i)$, where $\mathbf{p}$ is a vector that contains the 2-D implicit (cartesian) lines coordinates as x and y of the estimated Mn's position, and $\mathbf{p}_i$ is a vector that contains the 2-D cartesian coordinates x_i and y_i of the true positions of the Rns. Hence, we can identify the position of a Mn, as expressed in the following equation:

$$h_{ij}(\mathbf{p}) = \left((\mathbf{p} - \mathbf{p}_i)^2\right)^2 - \left(\left(\mathbf{p} - \mathbf{p}_j\right)^2\right)^2 - D_{ij}. \tag{32}$$

Thus, the objective function of $h_{ij}(p)$ is given as

$$H(\mathbf{p}) = \sum_{i=1}^{N} h_{ij}(\mathbf{p})^2. \tag{33}$$

TDoA, in comparison to other STPMs, is less vulnerable to obstacles under NLoS conditions. This is due to the way it measures signal propagation and its communicational architecture, where the Rns are always aware and interconnected with each other. TDoA's accuracy depends on the type of IPS application and its complexity. The obtained simulated result of the TDoA method is represented in Fig. 21.

In accordance with the simulated results given in Fig. 21, TDoA's LoS positioning accuracy is under 1.9 m, while in NLoS conditions its accuracy is below 3 m. Although the TDoA method provides remarkable positioning accuracy, especially in NLoS conditions, its implementation process is very complex due to internode SYNC.

2) Time of Flight: ToF, also known as the ToA, is UWB's asynchronous ranging scheme method, where each node uses its own timing information to measure the ranging signal instead of using global SYNC. It is a tag- or mobile-side power consumption approach that requires accurate SYNC between Rn and iMn to calculate the position of an iMn based on the propagation time between Rn and iMn. Therefore, each node has to measure an RTT to obtain the propagation time and the distance between Rn and iMn, consequently. The ranging procedure in an asynchronous ranging scheme is bidirectional or two-way and can use relatively simple circuitry without the need for a precise crystal oscillator [30].

The ToF method, however, provides precise positioning accuracy; the time SYNC between Rn and iMn is considered to be its main drawback. It uses the SDS-TWR approach between iMn and each Rn, where iMn requires three messages, e.g., the poll messages, the response message, and the final message per Rn, as illustrated in Fig. 22, to calculate the distance of an iMn. Thus, all Rns need $3 \times n$ number of messages in total for n number of Rns [3].

iMn initiates the process by sending a poll message, where the Rn records the arrival time of the poll message, which is denoted as T_{rp}. The Rn responds with a response message as T_{sr}, and the iMn records the response's arrival time as T_{rr}, then sends a final message as T_{sf} to the Rn as T_{rf}. This then computes the ToF based on these recorded timestamps. ToF's communication flow between Rn and iMn is given in Fig. 22.

Each Rn has now enough information to calculate its distance from iMn. Considering the given procedure in Fig. 22(b), the RTT of $T_{i(x)}$ and their reply times $T_{j(x)}$ can

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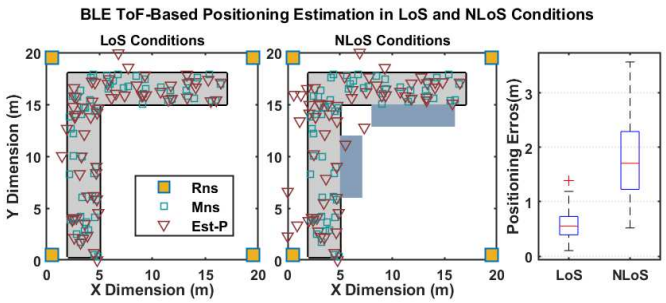


Fig. 23. UWB's ToF positioning results in LoS and NLoS conditions.

be expressed as

$$\begin{cases} T_{i(\mathrm{Mn})} = T_{\mathrm{rr}} - T_{\mathrm{sp}} \\ T_{i(\mathrm{Rn})} = T_{\mathrm{rf}} - T_{\mathrm{sr}} \\ T_{j(\mathrm{Mn})} = T_{\mathrm{sf}} - T_{\mathrm{rr}} \\ T_{j(\mathrm{Rn})} = T_{\mathrm{sr}} - T_{\mathrm{rp}}. \end{cases} \tag{34}$$

According to (34), we can compute the distance between iMn and Rns based on the time it takes for a signal to travel between them. The distance d can then be computed as $d = c \times t$ where d is the distance between iMn and Rn, c is the speed of light, and t is the time the signal takes to travel from iMn to Rn and back to iMn, which is the ToF RTT. Hence, ToF's two-way ranges can be computed as follows:

$$\mathrm{ToF} = \frac{\left(T_{i(\mathrm{Mn})} - T_{j(\mathrm{Rn})}\right) + \left(T_{i(\mathrm{Rn})} - T_{j(\mathrm{Mn})}\right)}{4} \cdot c \tag{35}$$

where T_{rr} is the time when Rn receives the signal from an iMn, T_{sp} is the time when iMn sends the signal, T_{tr} is the time when Rn sends a response back to iMn, and T_{rp} is the time when iMn receives the response.

ToF's positioning accuracy is higher in LoS conditions than TDoA due to direct measurements of the distance between iMn and Rn. While TDoA relies on the difference in arrival times, which can be affected by clock SYNC errors. On the other hand, ToF's accuracy under NLoS conditions is lower than TDoA, which is based on how each method measures and processes the signal data.

In NLoS scenarios, obstacles such as walls can cause signal delays and reduce the signal-to-noise ratio. This affects the accuracy of distance measurements in ToF. In addition, ToF relies on measuring the RTT of a signal between iMn and Rn, which requires precise SYNC and is more susceptible to errors caused by NLoS conditions, and the need for iMn to send and receive signals multiple times during each ranging process, which can be impacted by environmental factors. The obtained simulated results using the ToF method under NoS and NLoS conditions are represented in Fig. 23.

Although the obtained results in PBR, TDoA, and ToF are slightly different in LoS conditions, in the case of NLoS conditions, their accuracy degrades, with PBR being the most vulnerable method, followed by ToF and then TDoA, in the case of STPMs.

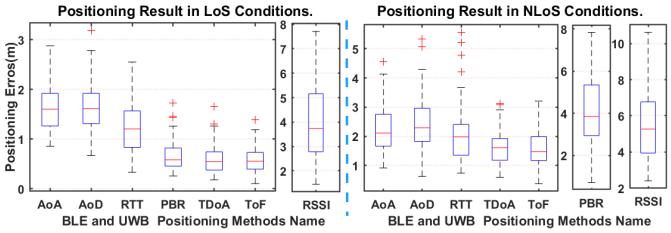


Fig. 24. BLE's and UWB's positioning results in LoS conditions.

D. Comparison of Positioning Methods in BLE and UWB

A general comparison of the evaluated methods (SSPMs, SAPMs, and STPMs) conducted in this article is made based on LoS and NLoS conditions. It has been observed that the positioning methods whose communication nature between the Rn and iMn is bidirectional are the most vulnerable under the NLoS conditions, while they offer higher positioning accuracy in LoS conditions.

The LoS and NLoS results in BLE's and UWB's positioning methods are given in Fig. 24, where it can be observed that RSSI has the highest positioning error, while ToF has achieved the highest accuracy based on the given scenario and parameters in the simulated environment under LoS conditions. On the other hand, in NLoS conditions, along with RSSI having high positioning error, which is due to multipath effects, the PBR's positioning error is higher than any of the SAPMs and STPMs. This is due to several factors, e.g., multipath effects, where signals bounce off obstacles, which leads to phase ambiguity, making it difficult to determine the true distance. Phase wrapping is another issue in NLoS for PBR since it relies on measuring phase shifts, where large phase variations due to reflections can cause incorrect distance calculations.

TDoA generally offers higher accuracy compared with ToF under NLoS conditions. This is because TDoA relies on the difference in arrival times at multiple Rns, which helps mitigate the impact of multipath effects and signal obstructions. While ToF is more susceptible to errors in NLoS environments due to signal reflections and delays, which can introduce significant bias in distance measurements.

The obtained results of each positioning method in this article are specific to BLE and UWB technologies in the simulated model. RSSI, AoA, AoD, RTT, and PBR methods are evaluated based on BLE technology, whereas the evaluation of the TDoA and ToF methods is based on UWB technology. The positioning results of BLE and UWB's methods in Fig. 24 consist of LoS and NLoS conditions in an indoor environment.

In addition, the cumulative distribution function (cdf) for each method in the LoS condition is given in Fig. 25, which further illustrates the differences between each method's results in BLE and UWB technologies, which represent the probability of their positioning errors under the LoS condition.

The cdf results of each method under NLoS conditions are given in Fig. 26, which represent the probability of their positioning errors. It further illustrates the differences between the obtained results in BLE's and UWB's positioning methods in NLoS conditions.

The evaluated results of BLE's and UWB's methods, as summarized in Table V, reveal a clear pattern and demonstrate

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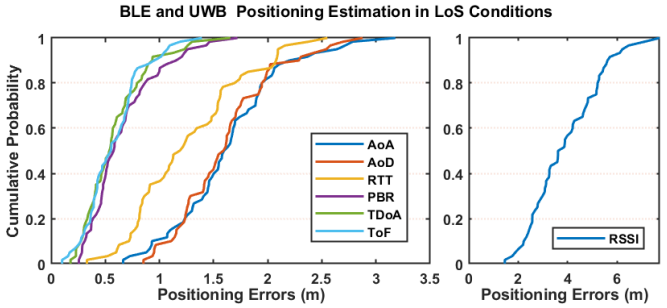


Fig. 25. CDF results of BLE's and UWB's methods in LoS conditions.

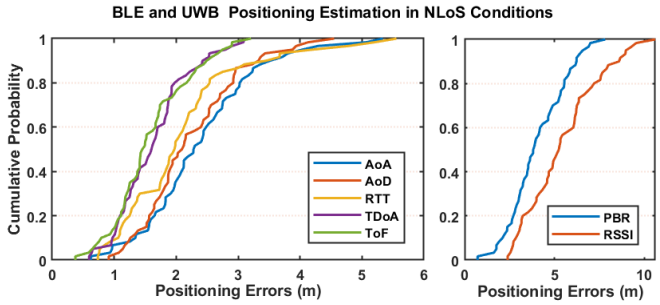


Fig. 26. CDF results of BLE's and UWB's methods in NLoS conditions.

TABLE V
BLE- AND UWB-BASED ACCURACY IN LOS AND NLOS CONDITIONS

Cond	RMSE	RSSI	AoA	AoD	RTT	PBR	TDoA	ToF
LoS	Max	**8.95m**	3.76m	3.40m	2.89m	2.75m	2.65m	2.64m
	Mean	4.22m	1.84m	1.72m	1.04m	0.84m	0.80m	**0.74m**
	Mode	1.28m	0.89m	0.66m	0.29m	0.17m	0.20m	**0.16m**
NLoS	Max	**11.62m**	5.33m	5.73m	5.85m	**8.28m**	3.89m	**4.19m**
	Mean	6.30m	2.61m	2.80m	2.50m	5.04m	2.08m	**2.00m**
	Mode	2.44m	1.02m	0.88m	0.86m	0.84m	0.73m	**0.41m**

varying levels of accuracy across the LoS and NLoS scenarios. Each method yielded distinct performance metrics, with some excelling in precision while others offered faster computation times or greater resilience to noise. For instance, ToF showed high accuracy in LoS environment, followed by TDoA, BPR, and RTT methods. In contrast, traditional methods like RSSI are considered less accurate. On the other hand, in NLoS conditions with the same simulation parameter but added obstacles (walls), the order of accuracy has changed. TDoA has the highest accuracy, followed by ToF, RTT, AoD, AoA, PBR, and RSSI. Overall, the comparison highlights trade-offs between positioning accuracies, offering valuable insights into each method for specific real-world applications.

The max, mean, and mode root mean square error (RMSE) results of BLE and UWB positioning methods, as detailed in Table V, exhibit substantial variability as a function of simulation parameter, considered noise, number of Rns and Mns deployed, and the obstacles in the indoor environment. These findings underscore the sensitivity of positioning accuracy to environmental and parametric variations, which we have considered in our simulation, suggesting that modifications to these parameters can lead to statistically significant changes in their performance outcomes.

TABLE VI
BLE- AND UWB-BASED POSITIONING METHODS

BLE-Based Positioning Methods			
Methods	**Security**	**Power Usage**	**Hardware**
RSS	Potential spoofing	Low	Basic BLE Device
AoA	Potential spoofing	Low	Beacon and Tag
AoD	Potential spoofing	Low	Beacon & Mobile node
PBR	Secured	Low	Initiator and Reflector
RTT	Secured	Low	Initiator and Reflector
UWB-Based Positioning Methods			
ToF	Secured	Moderate	Anchor and Tag
TDoA	Secured	Moderate	Anchor and Tag

Upon evaluating the primary positioning methods in BLE and UWB technologies, in contrast to their accuracy and behavior in an indoor environment. We have compared their power consumption, security, and the need for additional hardware, in their respective positioning methods, which is represented in Table VI.

Prior to examining the optimization strategies to enhance positioning precision in BLE and UWB technologies, it is essential to first comprehend the shared limitations these systems encounter under both LoS and NLoS conditions.

V. INDOOR POSITIONING CHALLENGES AND OPTIMIZATION STRATEGIES IN BLE AND UWB

In this section, we explored related positioning vulnerabilities of BLE's and UWB's methods in an indoor location, especially under NLoS conditions. Accordingly, we have proposed optimization filtering-based algorithms to optimize their positioning accuracy and performance in IPS applications.

A. Positioning Vulnerabilities in BLE and UWB Technologies

Almost all IPS technologies face obstacles in an indoor environment [3], [19]. This includes multipath radio propagation, signal attenuation, reflection in NLoS conditions, interference, and noise. However, the impact of these challenges varies based on the type of IPS technology [42], [71].

BLE and UWB pose several positioning vulnerabilities that compromise their positioning accuracy and security. While BLE is widely used due to its low-power consumption, it is susceptible to signal interference, multipath effects, and spoofing attacks, which are specific to BLE's older versions. UWB, on the other hand, is known for its high precision, but is vulnerable to early detection/late-commit (ED/LC) attacks, where adversaries manipulate ToA measurements to distort location data. In addition, both technologies can be affected by NLoS conditions, as we have witnessed in the comparison section. However, these vulnerabilities vary based on different positioning methods used in BLE and UWB technologies, e.g., UWB's positioning methods are proven to be less vulnerable to these challenges in comparison to BLE, while they remain highly susceptible to SYNC challenges [72].

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According to the classification of the positioning methods in this article, the SSPMs are highly affected in general, while the STPMs are categorized due to multipath propagation. This is caused due to the presence of obstacles in an indoor location, where signals bounce off walls and objects, creating multiple paths that distort the received signal strength or phase shift, making distance estimation inaccurate. Co-channel interference (CCI) is another major issue, especially in BLE technology, due to its shared frequency band with other wireless technologies [73]. CCI specifically occurs when two signals on the same channel compete with each other, leading to signal degradation and causing positioning error.

The SAPMs are vulnerable to reflection in NLoS conditions, e.g., indoor obstacles. It can significantly change the direction of signals, which leads to inaccurate angle measurements. They also require multiple antennas, which play a crucial role in signal transmission. However, a properly designed antenna array in consideration of environmental structure can overcome some of the posed challenges by offering low noise impact, strong signal strength due to beamforming, which leads to a better positioning accuracy in them [9]. On the other hand, designing a complex antenna array can increase the cost and complexity of implementation, although using the smartphone as a mobile device can illuminate part of the added cost, especially in the AoD method [29], [74].

And the STPMs, apart from being affected by multipath and NLoS issues [75], an accurate time SYNC between the Rn and Mn is even a major issue, which can be challenging to achieve in indoor environments. The SYNC problem varies and is specific to the utilization of STPMs, e.g., PBR and RTT require partial SYNC. ToF, on the other hand, requires a tight SYNC between Rn and iMn, and TDoA follows the concept of carrier SYNC or intertransmitter SYNC where the Mn needs to be synced with Rn, and the Rns between themselves. This makes a system very complicated.

PBR is vulnerable due to a number of different factors, such as multipath propagation, phase ambiguity, partial clock SYNC between iMn and Rn, e.g., coarse time SYNC (the initial time SYNC used to reduce timing offsets). This SYNC is important for phase measurement and frequency hopping (the ranging process occurs over multiple channels) to mitigate multipath fading, where the iMn and Rn perform carrier transmission and reception over the entire frequency band to improve the accuracy. Interference and noise are also part of the challenges in PBR.

RTT, on the other hand, is not only vulnerable to multipath propagation, phase ambiguity, clock SYNC, interference, and noise but also poses processing overhead, which processes the real-time timing data, and can be computationally costly. However, it is more resistant in NLoS and provides better positioning accuracy than the PBR method. BLE's STPMs, as of now, have only one communication scheme at a time, where the broadcast feature is not supported. This can increase measurement time and limit the scalability of an IPS application [42].

The TDoA method, apart from being vulnerable to NLoS conditions, multipath radio propagation, processing overhead, interference, and noise. It has a complicated communication flow due to its network SYNC, where Rns are not only synced with Mn but also between themselves. The parameters like rounding effect, outliers, filtering, packet loss detection, and thermal dependencies can affect the SYNC process in TDoA to a high level of inaccuracy [76].

In order to reduce the complexity of internode SYNC in TDoA, the ToF offers a different architecture where the Rns do not require SYNC between each other. It is only required between the Rn and iMn. This approach, however, reduces the complexity and internode SYNC between Rns. On the other hand, clock SYNC and processing overhead are the major issues in ToF alongside multipath propagation, phase ambiguity, interference, and noise. The clock SYNC between Rn and iMn can be very challenging, where any clock drift or offset can cause errors in time measurement. The ToF approach also requires high power in iMn to send and receive signals multiple times to each Rn, which leads to more power consumption in case the iMn is a battery-powered device [77].

In terms of scalability, ToF and TDoA methods can support unlimited Mn based on the positioning scheme. The AoA and AoD methods can support up to 1000 of Mns based on the positioning scheme as well as the RSSI, but on the other hand, BLE's PBR and RTT methods can support up to 10's of the iMn based on the positioning scheme due to device-to-device communication as of now [56].

Each method has its own potential, e.g., ToF does not require Rn to Rn SYNC, TDoA provides Rn awareness, reducing redundancy, AoA and AoD provide direction-finding features and offer long-distance communication services, and PBR and RTT provide more precise positioning accuracy and security in BLE technology. The communication range in BLE can reach up to 50 m, while BLE beacons can extend the range coverage up to 300 m in LoS conditions, and UWB ranges up to 50 m. With a UWB anchor, it can be extended up to 200 m.

B. Optimization Strategies for BLE and UWB Methods

The integration of filtering algorithms with BLEs and UWBs' positioning methods enhances their positioning estimation by systematically mitigating signal fluctuations and contextual interference. Algorithms such as kNN, Kalman filter (KF) [32], extended KF (EKF), unscented KF (UKF) [78], PF, support vector machine (SVM) [79], and Gaussian process (GP) [80] serve as probabilistic frameworks for the signal dynamics, compensating for multipath propagation and NLoS conditions in an indoor location. This is due to their continuous refinement of estimated position through recursive error correction and predictive modeling. These algorithms not only attenuate noise but also facilitate seamless indoor tracking even in environments with unstable signal conditions. The mutual interaction between filtering algorithms and BLE and UWB technologies signal analytics thereby represents a robust hybrid positioning approach, yielding enhanced reliability, adaptability, submeter accuracy in SAPMs, and centimeter

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accuracy in STPMs. Hence, it contributes to mitigating the signal reflection, attenuation, and related environmental noise.

Each of these algorithms has pros and cons, which are very specific to their use cases, such as linear and nonlinear-based applications. Their performance varies depending on the nature of a positioning method, e.g., in a nonlinear application like indoor navigation, using EKF, UKF, and PF can perform better due to their nonlinear approach. Although UKF and PF algorithms can offer better estimation results in a sophisticated system, on the other hand, they have higher computational complexity than EKF. This is due to the requirement of a large number of particles in PF for precise estimation, and UKF propagates multiple sigma points instead of linearizing with Jacobians as EKF [78].

It is essential to know the nature and behavior of a BLE- and UWB-based positioning method while integrating a filtering algorithm in order to obtain a reliable performance in an IPS application. Every classified portioning method employs a distinct nature of communication between Rn and Mn or iMn and Rn. Therefore, positioning methods within a given class should only be integrated with a filtering algorithm that optimally serves their communication needs. Otherwise, not only will their performance fail to improve, but their computational cost may also increase. Hence, the given pairs of each method with a filtering algorithm are based on their compatibility and their use case in an IPS application.

SSPMs can generally be optimized using kNN- and ML-based algorithms. They usually perform best with the Log-distance path-loss model, which can easily handle noise and is an environment-dependent method. For instance, ML models can learn complex signal patterns and outperform traditional models in cluttered spaces [80]. KF can be a good option for the RSSI method because it helps smooth out the high variance readings, which can improve stability and accuracy in RSSI [32]. On the other hand, SVM can be effective in fingerprinting-based systems for classifying locations based on signal strength [79].

SAPMs are primarily designed for nonlinear-based applications, such as indoor navigation and asset tracking in a warehouse. Based on the nature of SAPMs' communication, e.g., AoD and AoA, their selection of filtering algorithms for optimizing their positioning accuracy is somewhat limited and narrow. This is specific to algorithms offering inherently nonlinear dynamics, e.g., EKF, UKF, and PF [78]. These algorithms work by simulating many possible states (particles) and updating them based on likelihood, making them robust in complex environments, by handling nonlinearities and multimodal distributions well. UKF works better in handling nonlinearities compared with the EKF, but it is more computationally expensive [78]. Hence, the choice of these filtering algorithms depends on the type of an IPS application and its complexity.

Based on our evaluation of STPMs in BLE and UWB technologies, they have different principles of communication, e.g., ToF, TDoA, and PBR. Therefore, their utilization of filtering algorithms varies. For instance, PBR suffers from multipath effects and phase ambiguity, especially in NLoS, utilizing any of PF, EKF, or UKF algorithms can be an effective option in PBR, as they are particularly well-suited for nonlinear and non-Gaussian systems, which are common in the real world due to multipath effects and phase ambiguity [78].

ToF and RTT work well when we have synchronized clocks or can measure round-trip delays. The best way to optimize their positioning accuracy is to integrate them with the UKF algorithm, which is highly effective in handling the nonlinearities in time-based measurements compared with EKF, especially when the system model is complex [78].

In TDoA, one of the possible choices for better estimation is Chan's algorithm, and Taylor Series Expansion due to their robustness, which does not require initial guesses, and Taylor expansion is useful for iterative refinement [81]. Each positioning method relies on different signal properties; therefore, the filtering algorithms vary accordingly.

VI. Conclusion

In conclusion, the evolution of BLE and UWB technologies and their positioning methods reflects a paradigm shift in how IPS-based wireless communication technologies can simultaneously ensure accurate positioning estimation and location awareness. This article has evaluated BLE- and UWB-based positioning methods, which include RSSI, AoD, AoA, RTT, and PBR in BLE, ToF, and TDoA in UWB. BLE is a widely available technology with almost the same standards throughout the world, offering up to centimeter-level positioning accuracy in the utilization of CS techniques. On the other hand, UWB offers a wide frequency range with different guidelines specific to different regions. It provides higher positioning accuracy and security than BLE. However, BLE is a more energy- and cost-efficient technology which can be found in almost all modern devices such as smartphones, tablets, and laptops. However, challenges such as scalability, security, standardization demands, and handling NLoS conditions can be overwhelming in both technologies. UWB has low interference and high resistance to multipath effects, making it an ideal approach for IPS applications requiring precise location information in a complex environment. Meanwhile, BLE is favored for its lower power consumption and extended range. The choice between BLE and UWB technologies and their positioning methods depends on the specific requirements of an IPS application. Ultimately, this survey underscores the transformative potential of BLE and UWB in the IPS domain.

References

[1] C. Huang, Y. Zhuang, H. Liu, J. Li, and W. Wang, "A performance evaluation framework for direction finding using BLE AoA/AoD receivers," *IEEE Internet Things J.*, vol. 8, no. 5, pp. 3331–3345, Mar. 2021.

[2] Y. S. Khraisat, M. Al-Khateeb, Y. Abu-Alreesh, A. Ayyash, and O. Lahlouh, "GPS navigation and tracking device," *Int. J. Interact. Mobile Technol.*, vol. 5, no. 4, pp. 39–41, Oct. 2011.

[3] S. M. Asaad and H. S. Maghdid, "A comprehensive review of indoor/outdoor localization solutions in IoT era: Research challenges and future perspectives," *Comput. Netw.*, vol. 212, Jul. 2022, Art. no. 109041.

[4] P. S. Farahsari, A. Farahzadi, J. Rezazadeh, and A. Bagheri, "A survey on indoor positioning systems for IoT-based applications," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7680–7699, May 2022.

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[5] C. T. Li, J. C. P. Cheng, and K. Chen, "Top 10 technologies for indoor positioning on construction sites," *Autom. Construction*, vol. 118, Oct. 2020, Art. no. 103309.

[6] L. Botler, M. Spörk, K. Diwold, and K. Römer, "Direction finding with UWB and BLE: A comparative study," in *Proc. IEEE 17th Int. Conf. Mobile Ad Hoc Sensor Syst. (MASS)*, Dec. 2020, pp. 44–52.

[7] J. Singh, N. Tyagi, S. Singh, F. Ali, and D. Kwak, "A systematic review of contemporary indoor positioning systems: Taxonomy, techniques, and algorithms," *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34717–34733, Nov. 2024.

[8] M. R. Rashidi, C. Cherian, V. S. Negi, and S. Chinara, "IoT-enabled smart parking system using artificial intelligence and optical character recognition," in *Proc. 6th Int. Conf. Commun., Signal Process., Appl. (ICCSPA)*, Jul. 2024, pp. 1–6.

[9] S. Han, H. Yoo, H. Choo, and B.-J. Jang, "IEEE 802.15.4z UWB angle of departure tag design for indoor positioning," in *Proc. 53rd Eur. Microw. Conf. (EuMC)*, Sep. 2023, pp. 975–978.

[10] T.-M.-T. Dinh, N.-S. Duong, and K. Sandrasegaran, "Smartphone-based indoor positioning using BLE iBeacon and reliable lightweight fingerprint map," *IEEE Sensors J.*, vol. 20, no. 17, pp. 10283–10294, Sep. 2020.

[11] Y. Assayag, H. Oliveira, E. Souto, R. Barreto, and R. Pazzi, "Adaptive path loss model for BLE indoor positioning system," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12898–12907, Jul. 2023.

[12] R. Mazraani, M. Saez, L. Govoni, and D. Knobloch, "Experimental results of a combined TDOA/TOF technique for UWB based localization systems," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, May 2017, pp. 1043–1048.

[13] Y. Zhuang, C. Zhang, J. Huai, Y. Li, L. Chen, and R. Chen, "Bluetooth localization technology: Principles, applications, and future trends," *IEEE Internet Things J.*, vol. 9, no. 23, pp. 23506–23524, Dec. 2022.

[14] Decodio AG.(2023). *TDoA Localization: From Theory to the Field*. Accessed: Oct. 19, 2024. [Online]. Available: https://www.decodio.com/media/downloads/TDOA_Localization_From_Theory_to_the_Field_v1-0.pdf

[15] M. Kolakowski, "Improving accuracy and reliability of Bluetooth low-energy-based localization systems using proximity sensors," *Appl. Sci.*, vol. 9, no. 19, p. 4081, Sep. 2019.

[16] H. Ólafsdóttir, A. Ranganathan, and S. Čapkun, "On the security of carrier phase-based ranging," *IACR Cryptol. ePrint Arch.*, vol. 2017, p. 591, Apr. 2016.

[17] E. Bechthum et al., "30.6 A low-power BLE transceiver with support for phase-based ranging, featuring 5μs PLL locking time and 5.3 ms ranging time, enabled by staircase-chirp PLL with sticky-lock channel-switching," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 470–472.

[18] A. A. Morgan, "On the accuracy of BLE indoor localization systems: An assessment survey," *Comput. Electr. Eng.*, vol. 118, Sep. 2024, Art. no. 109455.

[19] J. Yang, C. Poellabauer, P. Mitra, and C. Neubecker, "Beyond beaconing: Emerging applications and challenges of BLE," *Ad Hoc Netw.*, vol. 97, Feb. 2020, Art. no. 102015.

[20] Z. Liu, T. Hakala, J. Hyyppä, A. Kukko, and R. Chen, "Performance comparison of UWB IEEE 802.15.4z and IEEE 802.15.4 in ranging, energy efficiency, and positioning," *IEEE Sensors J.*, vol. 24, no. 8, pp. 12481–12489, Apr. 2024.

[21] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 3rd Quart., 2019.

[22] S. J. Hayward, K. van Lopik, C. Hinde, and A. A. West, "A survey of indoor location technologies, techniques and applications in industry," *Internet Things*, vol. 20, Nov. 2022, Art. no. 100608.

[23] Y. Sartayeva, H. C. B. Chan, Y. H. Ho, and P. H. J. Chong, "A survey of indoor positioning systems based on a six-layer model," *Comput. Netw.*, vol. 237, Dec. 2023, Art. no. 110042.

[24] C. Gentner, P. Hager, and M. Ulmschneider, "Server based Bluetooth low energy (BLE) positioning using received signal strength (RSS) measurements," in *Proc. 13th Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Sep. 2023, pp. 1–7.

[25] N.-S. Duong and T.-M. Dinh, "Indoor localization with lightweight RSS fingerprint using BLE iBeacon on iOS platform," in *Proc. 19th Int. Symp. Commun. Inf. Technol. (ISCIT)*, Sep. 2019, pp. 91–95.

[26] S. Djosic, I. Stojanovic, M. Jovanovic, T. Nikolic, and G. L. Djordjevic, "Fingerprinting-assisted UWB-based localization technique for complex indoor environments," *Expert Syst. Appl.*, vol. 167, Apr. 2021, Art. no. 114188.

[27] J. Zou, Y. Sun, and Q. Wan, "An alternating minimization algorithm for 3-D target localization using 1-D AOA measurements," *IEEE Sensors Lett.*, vol. 4, no. 6, pp. 1–4, Jun. 2020.

[28] K. Hong, T. Wang, J. Liu, Y. Wang, and Y. Shen, "A learning-based AoA estimation method for device-free localization," *IEEE Commun. Lett.*, vol. 26, no. 6, pp. 1264–1267, Jun. 2022.

[29] W. Wang, B. Li, Q. Yin, and B. Yang, "A distributed localization in wireless sensor networks utilizing AOD estimation and rotation synthetic aperture technique," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2013, pp. 1474–1478.

[30] D. Wang, R. Kannan, L. Wei, and B. Tay, "Time of flight based two way ranging for real time locating systems," in *Proc. IEEE Conf. Robot., Autom. Mechatronics*, Jun. 2010, pp. 199–205.

[31] P. Zand, J. Romme, J. Govers, F. Pasveer, and G. Dolmans, "A high-accuracy phase-based ranging solution with Bluetooth low energy (BLE)," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2019, pp. 1–8.

[32] Q. Wang, Z. Li, H. Zhang, Y. Yang, and X. Meng, "An indoor UWB NLOS correction positioning method based on anchor LOS/NLOS map," *IEEE Sensors J.*, vol. 23, no. 24, pp. 30739–30750, Dec. 2023.

[33] Y. Chen, J. Wang, and J. Yang, "Exploiting anchor links for NLOS combating in UWB localization," *ACM Trans. Sensor Netw.*, vol. 20, no. 3, pp. 1–22, May 2024.

[34] J. Wang, Q. Gao, Y. Yu, X. Zhang, and X. Feng, "Time and energy efficient TOF-based device-free wireless localization," *IEEE Trans. Ind. Informat.*, vol. 12, no. 1, pp. 158–168, Feb. 2016.

[35] D. Coppens, A. Shahid, S. Lemey, B. Van Herbruggen, C. Marshall, and E. De Poorter, "An overview of UWB standards and organizations (IEEE 802.15.4, FiRa, Apple): Interoperability aspects and future research directions," *IEEE Access*, vol. 10, pp. 70219–70241, 2022.

[36] *Mesh Profile Specification: 1.0*, SIG Bluetooth, Kirkland, WA, USA, 2017.

[37] J. Tosi, F. Taffoni, M. Santacatterina, R. Sannino, and D. Formica, "Performance evaluation of Bluetooth low energy: A systematic review," *Sensors*, vol. 17, no. 12, p. 2898, Dec. 2017.

[38] F. Wu, C. Qiu, T. Wu, and M. R. Yuce, "Edge-based hybrid system implementation for long-range safety and healthcare IoT applications," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9970–9980, Jun. 2021.

[39] SIG Bluetooth, "Bluetooth core specification version 4.0," *Specification Bluetooth Syst.*, vol. 1, no. 7, p. 206, 2010.

[40] J. Yin, Z. Yang, H. Cao, T. Liu, Z. Zhou, and C. Wu, "A survey on Bluetooth 5.0 and mesh: New milestones of IoT," *ACM Trans. Sensor Netw.*, vol. 15, no. 3, pp. 1–29, May 2019.

[41] Core Specification Working Group. (Aug. 27, 2024). *BLE 6.0 Core Specification*. Accessed: Oct. 22, 2024. [Online]. Available: https://www.bluetooth.com/specifications/specs/core60-html/

[42] K. E. Jeon, J. She, P. Soonsawad, and P. C. Ng, "BLE beacons for Internet of Things applications: Survey, challenges, and opportunities," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 811–828, Apr. 2018.

[43] C. Gomez, J. Oller, and J. Paradells, "Overview and evaluation of Bluetooth low energy: An emerging low-power wireless technology," *Sensors*, vol. 12, no. 9, pp. 11734–11753, Aug. 2012.

[44] M. Woolley. (2020). *Bluetooth® Core Specification Version 5.1 Feature Overview*. Accessed: Mar. 22, 2024. [Online]. Available: https://www.bluetooth.com/bluetooth-resources/bluetooth-core-specification-v5-1-feature-overview/

[45] S.-H. Lee and Y.-H. Lee, "Adaptive frequency hopping for Bluetooth robust to WLAN interference," *IEEE Commun. Lett.*, vol. 13, no. 9, pp. 628–630, Sep. 2009.

[46] M. Woolley. (2020). *Understanding Reliability in Bluetooth® Technology*. Accessed: Feb. 20, 2024. [Online]. Available: https://www.bluetooth.com/bluetooth-resources/understanding-reliability-in-bluetooth-technology/

[47] J. R. Fernandes and D. Wentzloff, "Recent advances in IR-UWB transceivers: An overview," in *Proc. IEEE Int. Symp. Circuits Syst.*, Jun. 2010, pp. 3284–3287.

[48] E. Karapistoli, F.-N. Pavlidou, I. Gragopoulos, and I. Tsetsinas, "An overview of the IEEE 802.15.4a standard," *IEEE Commun. Mag.*, vol. 48, no. 1, pp. 47–53, Jan. 2010.

[49] *IEEE Standard for Low-Rate Wireless Networks*, IEEE Standard 802.15.4-2015, 2020, pp. 1–800.

[50] *IEEE Standard for Low-Rate Wireless Networks—Amendment 1: Enhanced Ultra Wideband (UWB) Physical Layers (PHYS) and Associated Ranging Techniques*, IEEE Standard 802.15.4z-2020, 2020, pp. 1–174.

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[51] B. Schleicher, S. Ahmed, A. Trasser, and H. Schumacher, "Multi-user capability of an FM IR-UWB system," in *Proc. IEEE Int. Conf. Wireless Inf. Technol. Syst.*, Aug. 2010, pp. 1–4.
[52] V. Niemelä, J. Haapola, M. Hämäläinen, and J. Iinatti, "An ultra wideband survey: Global regulations and impulse radio research based on standards," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 2, pp. 874–890, 2nd Quart., 2017.
[53] S. Aditya, A. F. Molisch, and H. M. Behairy, "A survey on the impact of multipath on wideband time-of-arrival based localization," *Proc. IEEE*, vol. 106, no. 7, pp. 1183–1203, Jul. 2018.
[54] M. Stocker, H. Brunner, M. Schuh, C. A. Boano, and K. Römer, "On the performance of IEEE 802.15.4z-compliant ultra-wideband devices," in *Proc. Workshop Benchmarking Cyber-Phys. Syst. Internet Things (CPS-IoTBench)*, May 2022, pp. 28–33.
[55] Y. Boussad, M. N. Mahfoudi, A. Legout, L. Lizzi, F. Ferrero, and W. Dabbous, "Evaluating smartphone accuracy for RSSI measurements," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.
[56] M. Woolley. (2024). *Bluetooth Channel Sounding Technical Overview*. Bluetooth SIG. Accessed: Mar. 11, 2025. [Online]. Available: https://www.bluetooth.com/channel-sounding-tech-overview/
[57] M. Dong, H. Li, R. Yin, Y. Qin, and Y. Hu, "Scalable asynchronous localization algorithm with mobility prediction for underwater wireless sensor networks," *Chaos, Solitons Fractals*, vol. 143, Feb. 2021, Art. no. 110588.
[58] F. Afroz, R. Subramanian, R. Heidary, K. Sandrasegaran, and S. Ahmed, "SINR, RSRP, RSSI and RSRQ measurements in long term evolution networks," *Int. J. Wireless Mobile Netw.*, vol. 7, no. 4, pp. 113–123, Aug. 2015.
[59] M. Phunthawornwong, E. Pengwang, and R. Silapunt, "Indoor location estimation of wireless devices using the log-distance path loss model," in *Proc. IEEE Region 10 Conf.*, Oct. 2018, pp. 499–502.
[60] D. J. Suroso, M. Arifin, and P. Cherntanomwong, "Distance-based indoor localization system utilizing general path loss model and rssi," *J. Robot. Control*, vol. 1, pp. 199–207, Nov. 2020.
[61] R. Wilfinger, T. Moder, M. Wieser, and B. Grosswindhager, "Indoor position determination using location fingerprinting and vehicle sensor data," in *Proc. Eur. Navig. Conf. (ENC)*, 2016, pp. 1–9.
[62] K. Xiao, F. Hao, W. Zhang, N. Li, and Y. Wang, "Research and implementation of indoor positioning algorithm based on Bluetooth 5.1 AOA and AOD," *Sensors*, vol. 24, no. 14, p. 4579, Jul. 2024.
[63] S. Ghosh and D. Sen, "An inclusive survey on array antenna design for millimeter-wave communications," *IEEE Access*, vol. 7, pp. 83137–83161, 2019.
[64] G. Pau, F. Arena, Y. E. Gebremariam, and I. You, "Bluetooth 5.1: An analysis of direction finding capability for high-precision location services," *Sensors*, vol. 21, no. 11, p. 3589, May 2021.
[65] Y. Shi. (2020). *Tests of Bluetooth Low Energy 5.1 Indoor Navigation—Direction Finding*. Accessed: Apr. 22, 2024. [Online]. Available: https://davidhoglund.typepad.com/files/gfm327_0e_bluetooth_low_energy_5.1.pdf
[66] A. Kranz. (2023). *Why Choose Bluetooth IPS Among the 8 Indoor Positioning System?*. Accessed: Mar. 19, 2024. [Online]. Available: https://www.mokoblue.com/bluetooth-indoor-positioning-system/
[67] M. Oshiro et al., "Sub-GHz phase-based ranging system: Implementation and evaluation," in *Proc. IEEE 91st Veh. Technol. Conf. (VTC-Spring)*, May 2020, pp. 1–7.
[68] N. Iwakiri and T. Kobayashi, "Joint TOA and AOA/AOD spectrum for ultra-wideband indoor double-directional channel estimation," in *Proc. IEEE 68th Veh. Technol. Conf.*, Sep. 2008, pp. 1–5.
[69] B. Xu, G. Sun, R. Yu, and Z. Yang, "High-accuracy TDOA-based localization without time synchronization," *IEEE Trans. Parallel Distrib. Syst.*, vol. 24, no. 8, pp. 1567–1576, Aug. 2013.
[70] M. Von Tschirschnitz and M. Wagner, "Synchronization-free and low power TDOA for radio based indoor positioning," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Sep. 2018, pp. 1–4.
[71] X. Li et al., "Wireless positioning: Technologies, applications, challenges, and future development trends," *Comput. Model. Eng. Sci.*, vol. 139, no. 2, pp. 1135–1166, 2024.
[72] P. Park, P. Di Marco, J. Nah, and C. Fischione, "Wireless avionics intracommunications: A survey of benefits, challenges, and solutions," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 7745–7767, May 2021.
[73] H. K. Ghafi, C. Spindelberger, and H. Arthaber, "Modeling of co-channel interference in Bluetooth low energy based on measurement data," *EURASIP J. Wireless Commun. Netw.*, vol. 2021, no. 1, p. 143, Dec. 2021.
[74] B. Hanssens et al., "An indoor localization technique based on ultra-wideband AoD/AoA/ToA estimation," in *Proc. IEEE Int. Symp. Antennas Propag. (APSURSI)*, Jun. 2016, pp. 1445–1446.
[75] S. G. Leitch et al., "On indoor localization using WiFi, BLE, UWB, and IMU technologies," *Sensors*, vol. 23, no. 20, p. 8598, Oct. 2023.
[76] F. Zhang, L. Yang, Y. Liu, Y. Ding, S.-H. Yang, and H. Li, "Design and implementation of real-time localization system (RTLS) based on UWB and TDoA algorithm," *Sensors*, vol. 22, no. 12, p. 4353, Jun. 2022.
[77] Z. Niu, H. Yang, L. Zhou, M. Farag Taha, Y. He, and Z. Qiu, "Deep learning-based ranging error mitigation method for UWB localization system in greenhouse," *Comput. Electron. Agricult.*, vol. 205, Feb. 2023, Art. no. 107573.
[78] M. Kolakowski, "Comparison of extended and unscented Kalman filters performance in a hybrid BLE-UWB localization system," in *Proc. 23rd Int. Microw. Radar Conf. (MIKON)*, 2020, pp. 122–126.
[79] C. H. Lam, P. C. Ng, and J. She, "Improved distance estimation with BLE beacon using Kalman filter and SVM," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2018, pp. 1–6.
[80] D. D. Nguyen and M. Thuy Le, "Enhanced indoor localization based BLE using Gaussian process regression and improved weighted kNN," *IEEE Access*, vol. 9, pp. 143795–143806, 2021.
[81] H. Chen, L. Cao, and Y. Yue, "TDOA/AOA hybrid localization based on improved dandelion optimization algorithm for mobile location estimation under NLOS simulation environment," *Wireless Pers. Commun.*, vol. 131, no. 4, pp. 2747–2772, Aug. 2023.



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